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MPP126 - Volvo Energy

Exterior Design of Battery Energy Storage Systems

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Abstract

This report outlines the design methodology, development, and engineering reasoning behind a project conducted in collaboration with Volvo Energy to create new exterior design concepts for Battery Energy Storage System (BESS) units. The project was conducted during spring 2026 by five students in the Master's Program at Chalmers University of Technology. Building directly upon prior market and user need analyses, the project progressed from establishing requirements specifications to broad ideation utilizing various creative methods. The team employed a morphological matrix for concept generation and systematic methods for concept evaluation, advancing through detailed concept development and high-fidelity CAD modeling and rendering to arrive at three distinct exterior directions: Sustainable, Industrial, and Scandinavian. The results provide a structured foundation for further development and highlight a clear opportunity for Volvo Energy to differentiate its BESS offerings through exterior design. By providing three distinct design bases tailored to fit into different environmental contexts, this work expands Volvo Energy's design research and future product architecture. Ultimately, the resulting final design concepts demonstrate a powerful potential to establish a new standard for charging infrastructure by integrating industrial engineering principles with a clear brand identity, these solutions position Volvo Energy to lead the industry not only through performance and technology, but through premium design and seamless environmental integration in a market where competitor products have little differentiation.

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1 Introduction

This section gives an overview of the project and its context. It begins with a brief background, which highlights the emerging potential found within Battery Energy Storage Systems (BESS) and the urgency to transform the design into a more modern solution. The aim of the project is then presented, defining the objectives of the project. Lastly, the scope is established to define the project boundaries.

1.1 Background

In today's rapidly evolving technological landscape, increasing electrification and growing sustainability demands are driving the need for smarter and more resilient energy solutions. In this landscape of technology, Battery Energy Storage Systems (BESS) have emerged, showing great potential for enhancing energy grid stability, enabling higher penetration of renewable energy and functioning as electric vehicle charging infrastructure (Arvidsson et al., 2025). BESS units have several applications for different sectors of society, which means that they will be placed in public places (Anastasovski et al., 2024). Therefore, there is a need for these units to exhibit well-developed exterior design, in addition to their essential technology and internal components.

1.2 Project Aim

The aim of the project was to re-design and re-imagine the exterior design of the Volvo Energy BESS unit into a discreet, clean and modern-looking energy storage solution that represents the Volvo brand and contributes positively to overall performance.

1.3 Scope

The scope of the project specifically focused on the exterior design of the Volvo Energy BESS unit, positioning its visual appearance as a primary concern alongside functionality. While overall technical performance remains essential for the product as a whole, this project emphasized exploring the visual aspects and optimizing the exterior to achieve a discreet, clean, and modern appearance that reflects the Volvo brand.

Moreover, functional and technical requirements need to be fulfilled to justify any exterior- or visual feature, and served to define the boundaries of the design space, ensuring alignment between visual and technical aspects. Consequently, the prioritization of visual aspects did not treat functionality and design as separate elements, but rather emphasized their integration. In this project, exterior design was approached as a way to present the product's capabilities in a visually appealing, practical, and cost-effective manner while also allowing for future adaptation and refinement. This approach was reflected in both the selection and application of methods, as well as in the decisions made throughout the development process.

1.4 AI Assessment

AI tools were used to improve the readability of the report through support with spelling, grammar, sentence structure, and phrasing. AI was also used to generate L^AT_EX code for implementation in Overleaf when compiling the final PDF version of the report. The reliability of this code was verified through successful compilation and review of the resulting document. In addition, AI tools were used to create visual material for presenting the final project outcome. These visualizations were generated based on the project team's own CAD models and design work. Their accuracy and visual quality were validated against the project requirements and the final design outcomes presented in the report. Consequently, all analysis, evaluations, design activities, and conclusions of the project were carried out independently by the project team. AI did not influence the project's methodology, findings, or decision-making, but rather supported the communication and presentation of the work conducted.

1.5 Reading Guide

Following this introduction, Section 2 establishes the company background, current product portfolio, and the organizational backing of the Volvo Group. Section 3 establishes the foundational market gap and societal motivations for engineering an advanced exterior aesthetic. Section 4 breaks down the regulatory environment, customer and user segmentation, and evaluates direct market competitors. Section 5 outlines the iterative concept generation phase, detailing the technical validations, patent landscape, and precise design specifications for the three distinct design paths: Industrial inspired, Scandinavian inspired, and Sustainable inspired. Physical realization of these layouts via rapid scaling and modeling is outlined in Section 6. Section 7 and Section 8 provide a robust commercial assessment, including a base-case financial model and sensitivity parameters, followed by a comprehensive roll-out and exploitation strategy. Section 9 presents a critical discussion regarding methodology validity, process adaptations, and implementation barriers. Finally, concrete design directions and next steps are formulated in Section 10 and Section 11, culminating in a project summary in Section 12. Supplementary technical sheets, competitor data, survey responses, and the individual reflection reports are cataloged in Appendices A through E.

2 The Company - Volvo Energy

This project was developed in collaboration with Volvo Energy, a business unit within the Volvo Group focused on energy storage and charging infrastructure.

2.1 Company Background

Founded in 1927 in Gothenburg (Volvo Energy, 2023a), the Volvo Group has evolved from a traditional automotive manufacturer to a global provider of transport and infrastructure solutions. In response to the shift toward electrification and renewable energy, Volvo Energy was established in 2021 as a dedicated business unit focused on energy storage, charging infrastructure, and battery lifecycle management. Volvo Energy headquarters is situated in Gothenburg, with 143 employees and operations at 5 locations (Volvo Energy, 2024b). The company aims to accelerate and ease the transition to electrification, operating through seven core focus areas:

1. Re-manufacture, refurbish, and recycle batteries
2. Repurpose batteries
3. Digitalize to decarbonize
4. Create partnerships across ecosystems
5. Monitor and optimize batteries
6. Secure charging infrastructure and related e-mob services
7. Live and lead the change

(Volvo Energy, 2023a)

2.2 Current Position and Product Portfolio

Volvo Energy is a relatively new but strategically important player in the growing energy market, offering solutions that support reliable, cost-efficient and renewable energy use. Overall, Volvo Energy aims to operate as a system-level provider rather than a single-product manufacturer.

The company's core offering consists of BESS, which enable users to store energy when supply is abundant and inexpensive, and use or sell it during peak demand. These systems help optimize energy costs, ensure operational continuity and support the integration of renewable energy. The BESS units consist of several key components working together: batteries for energy storage, an inverter for power conversion, battery and energy management systems for control and optimization and thermal systems to maintain safe operating conditions (Volvo Energy, 2025c). Volvo Energy currently offers two main BESS products (Volvo Energy, 2025a):

1. PU2000 (Stationary Power Unit):

A large-scale solution (1000 kW / 2000 kWh) designed for commercial and industrial applications, which is illustrated in Figure 1.



Figure 1: PU2000 (Volvo Energy, 2025a)

2. PU500 (Portable Power Unit):

A flexible, mobile solution (up to 540 kWh / 240 kW) designed for temporary or dynamic use cases, which is illustrated in Figure 2.



Figure 2: PU500 (Volvo Energy, 2025a)

In addition to BESS, Volvo Energy provides charging infrastructure for electric trucks and machinery, including depot charging solutions and the development of public charging networks through partnerships (Volvo Energy, 2025b).

2.3 Customers and Market Context

This section outlines the customer and market landscape, further elaborated in Chapter 4. Volvo Energy operates strictly in a Business-to-Business (B2B) model, targeting commercial and industrial actors that invest in energy infrastructure and require reliable, high-capacity solutions (Arvidsson et al., 2025).

Geographically, the primary market is Sweden and the broader European region, driven by proximity, a home-market advantage, and shared regulatory frameworks and standards. Other important drivers include the "EU battery passport" which favors re-purposed batteries (Jalthar, 2025), giving Volvo Energy a competitive edge in the European BESS market, as they are planning to utilize repurposed batteries in the future. Due to these reasons, the European market is currently the most viable, while global markets represent secondary opportunities for future expansion (Arvidsson et al., 2025).

The BESS market can be divided into the following four categories: utility scale, small-scale Commercial and Industrial (Small C&I), large-scale Commercial and Industrial (Large C&I) and residential. The differences between these are the energy storage capacity and the application (Spinella & Poglitsch, 2024). Volvo Energy's PU2000 is categorized as a standalone Large C&I system. Thus, the core customers of the company are organizations investing in Commercial & Industrial (C&I) systems within the charging or construction sector. In this B2B context, the customer is not always the end user, but their requirements strongly influence system design.

The growing demand for high-power charging along national highway infrastructure and in urban environments increases the need for BESS solutions that not only provide high functional performance but are also visually integrated, as these systems are becoming more visible and are often deployed in environments such as charging stations (Arvidsson et al., 2025).

This focus is supported by prior research by Arvidsson et al. (2025), where segments were evaluated based on BESS integration and aesthetic maturity, which is illustrated in Figure 3. The charging sector shows both relatively high integration and greater sensitivity to visual quality compared to other sectors.

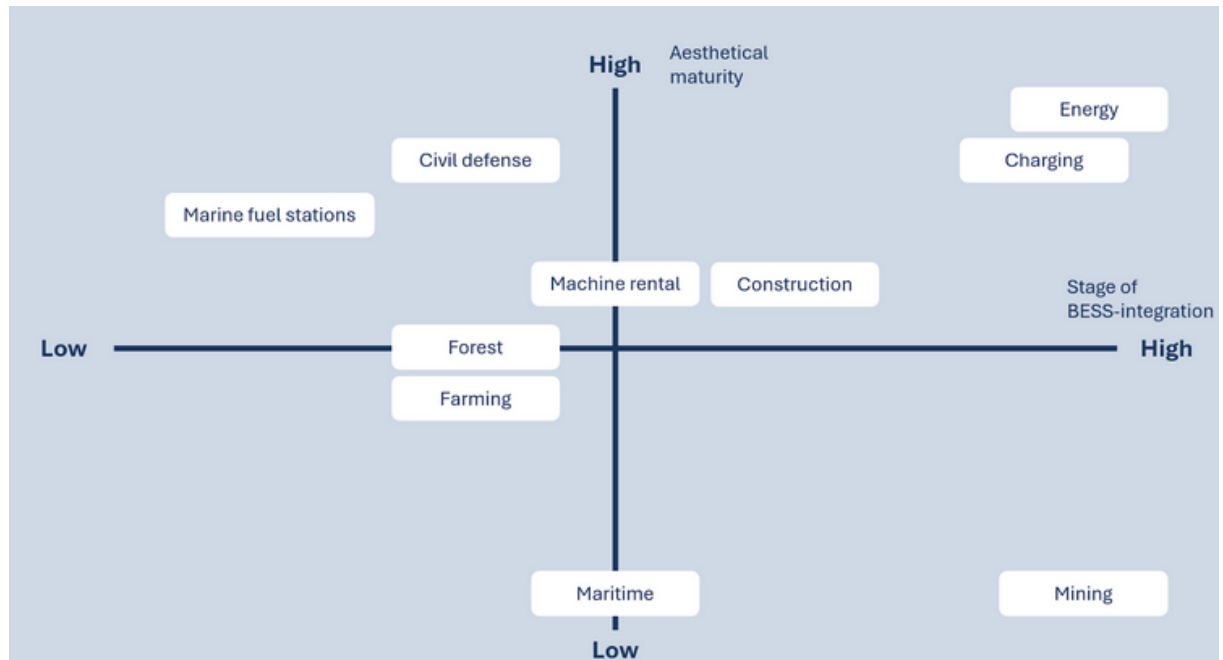


Figure 3: Quadrant analysis

2.4 Resources and Capabilities

Volvo Energy benefits from the extensive resources of Volvo Group, significantly strengthening its ability to develop and scale complex systems such as BESS. These resources can be structured as follows.

2.4.1 Personnel and Expertise

Volvo Energy has access to multidisciplinary engineering teams across mechanical, electrical, and software domains. This is complemented by strong competencies in safety-critical system design and extensive experience in large-scale industrial production.

2.4.2 Economic Resources

As part of a global corporation, Volvo Energy benefits from strong financial backing. This enables long-term investments in research and development (R & D), and infrastructure, which are essential in a capital-intensive and rapidly evolving industry such as energy storage. In addition, Volvo Group gives the company access to a strong existing customer base, providing a competitive advantage compared to less established companies within the industry.

2.4.3 Partners and Ecosystem

The company operates within a broad ecosystem of partners and a secure and stable supply chain of the Volvo Group, providing yet another competitive advantage. Moreover, newer strategic partnerships play a key role in accelerating Volvo Energy's mission. A notable example is Milence, a joint venture established in 2022 with Daimler Truck and Traton Group to develop high-performance public charging infrastructure across Europe (Volvo Energy, n.d.-a). In addition, Volvo Energy has invested in Connected Energy to advance second-life battery storage solutions and support circular energy systems (AB Volvo, 2022; Volvo Energy, 2023b). These collaborations reflect a broader strategy in which partnerships are used to share expertise, reduce risk and drive innovation.

2.4.4 Equipment and Infrastructure

Volvo Energy has access to advanced manufacturing and assembly facilities, as well as testing environments for safety, durability and performance validation. Their production site in Skövde supports the development and scaling of BESS solutions.

2.4.5 Organizational Structure

Operating as a focused business unit within a large corporation, Volvo Energy combines agility with industrial reliability. This structure enables faster innovation and development while maintaining the standards, processes, and stability associated with an established global company. Moreover, the established success of Volvo Cars and Volvo Group means that the Volvo brand name carries a significant advantage, as it is strongly associated with safety, quality, and stability.

2.5 Future Plans and Strategic Ambitions

Volvo Energy, as part of Volvo Group, aims to support the transition toward a fully electrified and fossil-free future. Their strategy focuses on enabling carbonization of transport and industry through integrated solutions combining vehicles, charging infrastructure, energy storage and digital services (Volvo Energy, 2024b).

A key priority is scaling BESS to support reliable energy supply, cost optimization and renewable energy integration. In parallel, Volvo Energy is developing circular battery value chains, including second-life applications, where used vehicle batteries are repurposed for stationary storage, contributing to the goal of net-zero emissions by year 2040 (Volvo Energy, 2024a).

Moreover, the company plays an active role in expanding public charging infrastructure for heavy-duty vehicles, particularly along key transport routes, supported by partnerships and digital platforms that improve accessibility and reliability (Volvo Energy, n.d.-b).

Within this context, the project addresses the growing importance of BESS exterior design. As these products become more visible in public and semi-public environments, their exterior evolves from needing to adhere to only functional requirements, to becoming a key interface between system and surroundings. Therefore, the project aim of transforming BESS from simple containerized units into clean, modern infrastructure elements that reflect the Volvo brand, while meeting technical, safety, and operational requirements is aligned with current market conditions and anticipated future developments in the BESS sector.

2.6 Likelihood of Project Success

The development of a new exterior design for BESS aligns closely with Volvo Energy’s strategic direction and market evolution. As energy storage systems become more integrated into urban- and industrial landscapes, their design becomes increasingly important.

Several factors support the successful implementation of this project. Mainly, Volvo Energy’s position within Volvo Group provides strong backing and access to extensive engineering resources, enabling investment in design innovation while ensuring feasibility, safety and performance are maintained. This combination of economic strength and technical expertise reduces risk in developing and implementing new design solutions. At the same time, the established Volvo brand offers a clear set of values, such as safety, quality and reliability, that can be translated into the product’s exterior design. Together with the growing demand for electrification, this creates a strong market pull for scalable, user-friendly and well-integrated solutions. Within this context, design differentiation becomes a key opportunity: most existing BESS solutions are standardized and container-like, often lacking visual integration. This creates clear potential for Volvo Energy to distinguish itself through a more refined and context-aware design approach.

At the same time, main challenges are balancing cost efficiency, standardization and manufacturability with innovation in design and user experience. Exterior design must operate within strict constraints related to logistics, thermal performance and regulatory compliance. Therefore, the success of this project depends on achieving a precise integration of aesthetic elements with technical- and economic feasibility.

3 The Need for a New Design

Currently, as demonstrated in Appendix A, a common theme across all competitors' exterior designs is the use of white, shipping-container-based enclosures. This indicates a gap in exterior design, presenting a significant opportunity for differentiation. The exterior of the BESS functions not only as a protective enclosure for the BESS, but also as a medium through which Volvo Energy can visually communicate safety, quality, and reliability. As the company strives to expand within the energy sector and public charging infrastructure, the visibility of the system in everyday environments increases. Consequently, the importance of exterior design has increased, transforming it from a secondary consideration into a strategic element.

Based on the previous section, the customer needs targeted with a new design of the BESS unit concern the general expression, public perception, and overall integration of the BESS unit with its surrounding environment.

A new exterior design aims to reduce the industrial complexity often associated with a product like the BESS unit. When placed in a public environment with frequent interactions from passersby, the current design found on the market can be perceived as rather intimidating or alarming. Additionally, a new exterior design strives to transform the BESS unit from a purely functional product into an element that harmonizes the system with its surrounding environment.

By changing the exterior design, two primary societal needs are addressed. The first need is concern regarding public trust and safety. By integrating a large-scale battery storage system like a BESS, concerns regarding safety can often arise. These concerns can be reduced by visually applying the core values of safety and quality to the exterior design, by building public confidence in the product. The second need is the concern regarding urban harmony and seamless integration into multiple user environments. As urban architecture and modern cities constantly evolve, the infrastructure and electrical grid need to keep up with this evolution. By creating a seamless and easily integrated exterior design, it can help contribute to the overall environment rather than distract.

4 Market Analysis

This section presents a market analysis of the BESS industry with focus on market structure, stakeholders, competitors, and product requirements. The analysis further explores opportunities for differentiation through exterior design within an increasingly standardized market.

4.1 Customers

The customers of BESS, purchasing and owning the systems, were identified as companies investing in C&I systems, commercial site owners, and charging infrastructure. Since the customer usually solely owns or purchases the system and the business is primarily B2B, the customer and primary user are not necessarily the same. The customers investing in the product results in their requirements and needs affecting the exterior design of the BESS.

4.2 Users

This subsection defines the different user segments and how these users are likely to interact with the product.

Primary Users

Primary users are actors who manage and depend on the battery energy storage systems for their business. Primary users include industrial site owners such as manufacturing plants and warehouses, commercial vehicle operators such as bus depots and logistics companies, and providers of fuel and mobility services including fuel stations. Primary users interact directly with the BESS through regular operation which makes the design of the exterior crucial for accessibility and safety.

Secondary Users

Secondary users are actors who maintain and operate the BESS. The secondary users include installers, maintenance and service technicians as well as system integrators such as electricians. These users are, like primary users, interacting with the BESS which makes them affected by the exterior design in terms of its installation efficiency, safety and serviceability.

Tertiary Users

Tertiary users are actors who are affected by the early life-cycle stages of the BESS. These users include assembly and manufacturing personnel whose work is affected by the exterior design of the BESS in terms of efficiency, safety and production costs.

Lead Users

Lead users are actors whose needs are recognized earlier than the needs of the average market. The lead users for the systems are fuel and mobility service companies providing electrical vehicle (EV) charging. These users might have needs that displays earlier than others since they are one of the first commercial actors to use EV charging at scale. For these users, both functional performance and aesthetics of the exterior design are central since they operate in public environments.

Extreme Users

Extreme users are actors whose needs many times exceed those of the average user. These users are those with unusually high power or aesthetic demands. However, due to such users being currently limited or not yet well established on the present market, they are not further explored within this project.

4.3 Stakeholders

Stakeholders are actors who, without using or purchasing it, are affected by the operation, development or deployment of the product. Stakeholders for the BESS include passerby, grid operators, regulators, local communities, company owners or decision-makers at companies using BESS, research and knowledge institutions and Volvo Energy as a company.

4.4 Customer Segments

Customer segment defines one or several groups of customers a company aims to reach and serve. Volvo Energy sells strictly B2B and the main focus regarding business areas in this project were identified as charging infrastructure and energy grid operators.

4.5 Requirements

Bess standards

There are a set of certain criteria that must be met by containerized BESS in order for it to be legal. This section covers standards that are within the scope of this project and are influencing the exterior design.

Safety, Fire and Explosion

Standards within these categories aim to reduce risks, including thermal runaway, electrical faults and build up of flammable gases and they influence the exterior design in multiple ways. One way the exterior design is impacted by these standards is by determining the design of deflagration venting.

Ingress Protection

Ingress Protection (IP) standards specify the exterior enclosure’s effectiveness against water and dust. These standards affect the design of the exterior by affecting protective elements such as gaskets and seals and are thereby essential for ensuring reliability.

Transportation and Labeling

These standards cover the legal and logistical aspects of transporting BESS units. This affects the exterior design by requiring it to withstand shocks, vibrations and varying climate conditions. Several types of hazard labels must also be included on the exterior and therefore taken into consideration in the design process.

4.6 Competitors

The global and European BESS market includes a wide range of competitors providing solutions within the segments of Residential, Small C&I, Large C&I, and Utility-scale. Companies offering BESS products in the large-scale C&I segment, the Utility-scale segment or both are considered Volvo Energy’s closest competitors as they function within the same market and are therefore likely to be most affected by a potential new solution. Mentioned below are direct competitors to Volvo Energy, offering similar energy storage systems.

Alfen

Alfen is a Dutch company specializing in energy solutions and operating across the European market (Arvidsson et al., 2025). Their BESS product called “The Battery”, see Appendix A Figure 45, resembles the Volvo Energy PU2000. The Battery is a stationary 1000-2000 kWh BESS confined within a transportable ISO container. Alfen currently has 132 energy storage projects and is present in over nine European countries. The company holds a strong market position in the Netherlands and reports a fast international growth as a result of its fast mover advantage (Arvidsson et al., 2025).

Wärtsilä

Wärtsilä is a Finnish company operating in 77 countries globally and has more than 19 GWh of installed energy storage capacity across over 130 projects. The company positions itself as an important player in the energy storage sector and with products such as the Quantum HE, which is a modular and compact energy storage system, the company is a key competitor to Volvo Energy (Arvidsson et al., 2025).

Eaton Corporation

The American-Irish power management company Eaton Corporation is present in over 160 countries and offers a range of different BESS solutions on the European market. Eaton’s X-storage container provides a similar capacity to Volvo Energy’s PU2000, positioning Eaton as a key competitor (Arvidsson et al., 2025).

4.6.1 Competitor Analysis

According to Arvidsson et al. (2025), the majority of firms currently present on the BESS market are well-established corporations, or subsidiaries with presence in several segments of the global energy market. These companies have the capacity to offer a diverse range of products where BESS is one of several focus areas. This shows how the market is mainly controlled by larger, established companies with widespread resources.

4.7 Competing Products

This subsection provides a description of what competing products offer in comparison to what is offered by Volvo Energy's current solution.

4.7.1 Exterior Design

When comparing exterior design performance between different companies providing similar BESS solutions to Volvo Energy, clear industry standards are shown. The properties compared are Ingress Protection (IP) classes, corrosion resistance, noise emission levels, and operable temperature and humidity ranges. For the systems compared the IP ratings are similar and the corrosion resistance is between C3 and C5. Noise emission for most systems ranges from 70 and 75 dBA. Most of the compared systems operable temperature ranges from -30 and 55 °C. All systems are painted white for heat reflection to optimize battery performance. The majority of systems handle at least 95% humidity. Some systems only require access from one side for maintenance and service, while others require access from multiple sides (Arvidsson et al., 2025).

4.7.2 Cost Structure

Without access to the exact price for the systems analyzed some sources suggest that the price ranges between 500 and 7000 SEK per kWh of usable capacity. This means that for a storage capacity offered in the large C&I segment, 200 kWh, the price is between 7-14 MSEK. The primary drivers of cost for BESS are the battery modules whereas the exterior design minimally affects the cost (Arvidsson et al., 2025).

4.7.3 Safety Solutions

Comparing safety solutions used for the different systems highlights that BESS utilizes a multi-layered defense strategy to prevent events such as thermal runaway. The benchmarking identified different internal, detection, suppression, and structural solutions used by the companies apart from complying with standards, rules, and regulations (Arvidsson et al., 2025).

4.7.4 Aesthetics

When comparing the aesthetic aspect of the exterior design between different systems, it is noticeable that almost all of them look highly similar. As previously mentioned, all systems compared are painted white for heat reflection. Other aesthetic parameters such as shape, doors, and handles are also similar. Generally the systems are modified containers with some more modifications than others. Volvo Energy's PU2000, see Appendix A Figure 42, is highly similar to a standardized container with corner castings and forklift pockets. Tesla's system Megapack 2XL, see Appendix A Figure 35, shows a more minimalist exterior design, moving away from the more traditional container and shows resemblance with an electrical cabinet. Most large and permanently installed systems have a lower focus on aesthetics and are instead usually compared through performance and cost. For a more detailed benchmarking analysis see Arvidsson et al. (2025). Figures of all main competitor solutions can be found in Appendix A.

4.8 Opportunities

Scalability

Aligning the BESS capacity with the energy demand was found to be one of the most crucial factors when selecting a BESS based on previous research (Arvidsson et al., 2025). This factor implies trade-offs between cost, weight and energy capacity which created a potential opportunity in expanding market

reach by developing a scalable solution specifically designed to match the customer’s needs which means a scalable exterior design is needed.

Service and Maintenance

Service and maintenance on most existing systems require several access points. This limited the positioning of the system, but also created an opportunity to improve overall convenience for access.

Temperature Management and Safety

Most competitors use separate systems for temperature control and safety such as water sprinklers, liquid cooling, insulation, emergency stops as well as smoke and heat sensors (Arvidsson et al., 2025). Here an opportunity was identified in using the exterior design in supporting safety and temperature management, which could enhance performance, reduce weight and lower costs.

Burglary Protection and Anti-vandal Design

Burglary protection is of high importance for the BESS units, especially for units placed in public environments with limited security measures. Thereby an opportunity to incorporate anti-vandal features through exterior design was identified.

Aesthetic Exterior Design Focus

As previously mentioned, most systems comparable to the PU2000 have a lower focus on aesthetics. Customer analysis showed that the changing segment prefers an alternative design (Arvidsson et al., 2025). Combined with the limited options currently available, this reinforced the main focus of developing the exterior design and generated an opportunity to create a product that differentiates itself from the general market of BESS.

Sustainable Exterior Design Numerous potential customers invest in BESS as part of a green transition strategy, making sustainability highly relevant. Despite this, competitors rarely emphasize a comprehensive sustainability approach that includes exterior design (Arvidsson et al., 2025). This created an opportunity to differentiate the exterior design through a stronger and more holistic sustainability focus.

4.9 Market Channels

Market channels are paths through which a product or service reaches its customers, including intermediaries as well as methods for selling, delivering, and communicating value (Sheth & Sisodia, 2006). Due to Volvo Energy’s BESS offering being in an early stage, publicly available information on distribution channels is limited and was not considered within the scope of this project.

4.10 Sub-Suppliers

Since the information on Volvo Energy’s suppliers for BESS is also limited and outside of the scope for this project, no further analysis on sub-suppliers was made. Assumptions regarding parameters such as costs were based on the knowledge that Volvo Energy orders their modified containers for PU2000 from southern Europe.

4.11 Policy

According to a PESTEL analysis on Volvo Energy’s BESS conducted by Arvidsson et al. (2025), several policy initiatives and regulatory actions are shaping the market of energy storage on a global scale.

A trend shaping the BESS market is governments increasingly introducing national energy storage targets. In the United States, the Energy Storage Grand Challenge, a program to expand energy storage technologies and reduce the cost of storage, was launched in 2020 and received new funding while continuing to be expanded over time (Arvidsson et al., 2025). This shows the ongoing political involvement in the BESS market of energy storage, highlighting how policies continue to drive the global market by indicating long-term demand for energy storage solutions.

4.12 Global Alignment

This commitment and interest in energy storage solutions is present in multiple major economies, showing alignment globally in the energy transition. In both China and Germany energy targets and aims

regarding increased storage deployments have been set (Arvidsson et al., 2025). This highlights how political decisions are increasingly affecting the BESS market. It also strengthens the perception of energy storage and BESS as a long-term, critical component infrastructure rather than a short-lived technology trend. Global alignment, commitment and recognition from governments towards energy storage supports market growth and further development of BESS as a product both internally and externally.

4.13 Social Resistance

Despite strong policy support globally, the deployment of utility-scale BESS installations is constrained by local opposition and safety concerns. In several states in the US, community opposition regarding safety and fire risks has led to delays and cancellations of BESS projects (Arvidsson et al., 2025). This exhibits challenges within the market, where deployment is affected by societal acceptance and trust. Therefore managing safety and community engagement becomes increasingly important for expanding the BESS market.

4.14 Concluding Market Requirements

Based on the market analysis, several key requirements for BESS solutions have been identified. The market for BESS is highly regulated, which means that compliance with fire protection, safety, IP among other regulations are baseline requirements for all products on the market. The technical performance for BESS being standardized leads to competitiveness mainly being driven by efficiency, adaptation and scalability. However, this conformity of the available products on the market creates opportunity for differentiation through visual design. The market is shaped globally by regulatory actions and policy initiatives where the interest and engagement in BESS is rapidly growing. This suggests that BESS solutions will continue to be required to meet technical and regulatory demands as well as expectations related to usability, trust and societal acceptance.

5 Product Concepts

This section outlines the concept development process throughout the project. It begins with an overview of the product development process, including concept generation and evaluation, and continues with a presentation of the final concepts, along with their technical validation, patent considerations, and manufacturing adaptability.

5.1 Process

The process of developing the final concepts followed a structured development approach with frequent iterations. As knowledge gaps were continually filled, the process was adapted to integrate these new insights.

The nature of the project aim dictated a high prioritization of visual aesthetics, which created a deviation from traditional development processes that mainly prioritize functional utility. Consequently, the traditional dominance of functional requirements was mediated by the need to prioritize the product's visual appeal and refine the aesthetic character of the BESS, creating a unique tension between engineering constraints and visual design goals. The re-prioritization forced a departure from the standard process, placing visual identity at the forefront of the early development stages.

The digital tool Miro was used throughout the entire project as a platform for communication, creativity and for documenting the process.

5.1.1 Requirements and Initial Work

The development process was initiated with a creative workshop, during which the team collaborated with other product development teams to discuss and ideate based on the project aim. The workshop started with a brief introduction of the project aim and scope and was followed by discussion and ideation together with the other teams. Several questions addressing points of interest in the project had been prepared and guided the discussions. Some of the questions were:

- " How is the current exterior design interpreted? Why? "
- " What should energy infrastructure look like in a modern society? "
- " How could vents, doors,... become design features instead of disruptions? "

As this workshop was done early in the process, the questions were formulated in an open manner to allow for lively discussion and encourage exploring the whole design space. The input from the workshop was documented in Miro with notes of quotes and input from the students, along with manual sketches that were done in the workshop.

Furthermore, the process continued with creating and establishing a requirements specification and a mood board. The requirements specification included both requirements and desires (See XXX), addressing both functional and visual aspects. Identified customer needs along with data from Volvo Energy were the main sources of information for the specification. The translation of customer needs into measurable requirements ensured continuous alignment in the development process, as development decisions remained anchored in user expectations. By anchoring development decisions in these specific criteria, the process remained focused on fulfilling both the functional and aesthetic expectations of the customer. In the specification, the customer needs associated with the requirement was stated, along with which method would be used to test the fulfillment of the requirement.

Along with the requirements specification, a mood board was created, to establish a clear visual direction for the design. All team members first created their own mood boards, all of which were then reviewed, discussed, and combined into one, final mood board. In addition, several guiding words were decided upon and placed next to the mood board.

Creating both the requirements specification and a mood board with guiding words allowed for the process to stay focused on the design aesthetic, while still sufficiently addressing the functional requirements that the exterior design should enable and adapt to.

5.1.2 Idea Generation

During the idea generation stage, multiple methods were applied to effectively capture as many ideas and explore as much of the design space as possible. Input from the creative workshop along with the requirements specification, mood board and guiding words served as points of reference during for this stage.

Brainwriting was used as the initial method for ideation, as it has been shown to encourage and enhance group creativity, which was important during this phase of the project, to allow for creativity and free flow of ideas. The session was started by defining the design focus of the session, which was done by formulating a prompt, from which all ideation was going to be based on: *"How can the functional and safe BESS system be visually integrated and pleasing into its environment?"*. By sketching and writing in silence and several rounds were conducted as the sketches were passed around in the team, allowing for each team member to build on each other's sketches. All ideas were then discussed, clustered and sorted into categories. The process then proceeded by further evaluating the most promising sketches/concepts, using Scamper, to support further creativity and idea generation of high originality. These ideas were discussed according to the SCAMPER questions, and then further visualized with sketching according to what had been discussed.

In this phase, the morphological matrix was gradually populated, as more and more ideas were generated. The matrix included categories for both functionality and aesthetics. Additionally, ideas for different subfunctions that were found online were also added. Idea generation then served as the basis for concept generation. The combination of idea generation and continuously filling in the morphological matrix allowed for discussions and questions to arise, which could be investigated and resolved, leading to increased knowledge of the product and it's systems.

5.1.3 Concept Generation and Evaluation

During this phase of development, concept generation and elimination were alternated, with multiple iterations.

During idea generation, a large number of ideas was generated and added to the morphological matrix, which resulted in a significantly large matrix. In order to effectively prioritize the most important subfunctions, the matrix was divided into two, where the first one included key subfunctions that were essential for the BESS to function properly. In the second matrix, all subfunctions addressing less crucial subfunctions were included.

The first matrix, with the most essential categories, was used to generate 10 concepts. Each concept was generated according to a different themes, such as "Scandinavian", "Sustainability" and "Volvo Aesthetic". Each concept was visualized using sketching and then evaluated by the team. Each concept was discussed to thoroughly evaluate and then eliminate some of them. The discussion focused on addressing the benefits and drawbacks of each concept in relation to the requirements specification, moodboard, input from Volvo Energy and the knowledge that had been built up during the process. The benefits and drawbacks for each concept were documented on Miro, along with features or issues that needed further evaluation or clarification. Moreover, different ways of combining the strongest features from each concept was addressed. The discussion resulted in elimination of five concepts, leaving five concepts to go. During the discussion, it had also been identified that two the concepts would be stronger combined as one. After this was done, four concepts remained, which had the themes *Scandinavian*, *Sustainable*, *Industrial* and *Sci-Fi*.

The process proceeded to the second round of concept generation. Here, the themes and main ideas from the five generated concepts were kept, but using a more minimal approach, as suggested by Volvo Energy. During the first session of concept generation, each concept was formed by combining suitable features from each subcategory of the morphological matrix. This time, the same themes for each concept was kept but every concept was generated using the same blank canvas of a standard shipping container (which is the traditional shape for the BESS) and only essential features were added. At this point in the process, increased knowledge had also been gained regarding standards for shipping containers and their transportation, and the conclusion had been drawn that the most feasible approach was to keep a standard shipping container as the standard shape of the BESS, to allow for realistic implementation of the design in the future, for conforming with international standards for shipping and create a realistic way for manufacturing and assembly further along. This session of concept generation resulted in four

updated version of the original, themed concepts, which were further visualized using physical and digital sketching.

Moving on to further elimination, a Pugh-matrix and the VALUE-method were used to assess and evaluate concepts, and guide further elimination. The Pugh-matrix was applied first, comparing each of the four concept to Volvo's current BESS, the PU2000. The Pugh-matrix provided a clear ranking of which two concepts were the strongest, compared to the PU2000, which were the *Scandinavian* and the *Sustainable* concepts. However, the two weaker ones, *Sci-fi* and *Industrial* received the same low score.

The Pugh-matrix was effective for addressing some of the functional requirements for the concepts and for identifying the strongest two concepts among the four (Ulrich et al., 2020, p. 156). However, it did not address the aesthetic requirements of more subjective nature in a satisfactory manner. To address this and enable further elimination, the VALUE-method was used Wikberg Nilsson et al. (2021, p. 234). As the advantages, limitations and unique elements were discussed for each concepts, it was clear that the PUGH-matrix had successfully identified the two strongest concepts, because both methods were aligned in identifying the same two concepts as the strongest. Moreover, this method provided added clarity in the ranking of the two weaker concepts. As the *Sci-fi* concept had significantly more limitations than the *Industrial*, it was eliminated.

Moreover, during the development process, two input and feedback sessions were conducted with Volvo Energy and the Volvo Trucks design team, respectively. These sessions provided valuable insights and guidance for the project team, contributing to clearer direction in decision-making and prioritization. In each session, the team presented the current development status, after which engineers from Volvo Energy and designers from Volvo Trucks provided feedback and engaged in discussions that enhanced understanding of both the business context and the product requirements.

One key insight from these sessions was the importance of applying a degree of minimalism in the design. In the early stages of development, the team had used a structured approach for idea generation, to explore a wide range of ideas and enable exploring of the whole design space (Ulrich & Eppinger, 2011, p. 144). While this approach was effective in identifying different opportunities for design and functionality, it also became evident that BESS units are primarily expected to integrate into their surroundings rather than function as visually dominant elements. This input was therefore incorporated into the continued development process.

To summarize, the complete process of concept generation- and elimination resulted in 3 distinct concepts remaining; *Scandinavian*, *Sustainable* and *Industrial*. The three concepts were then visualized in greater detail and with the same level of development in digital sketching, to allow for a fair evaluation and comparison of the three concepts moving forward. In addition, quick and dirty prototyping was used, as three down-scaled shipping containers were 3D-printed and clothed in printed versions of each exterior design of each concept.

5.1.4 Concept Development

Following the initial concept generation and elimination phases, three distinct concepts were further detailed: Sustainable, Industrial, and Scandinavian. The focus of the concept development phase was to transition these high-level ideas into realistic, technically viable designs by establishing concrete specifications.

This phase began with an intensive, full-day collaborative workshop. The team systematically went through each concept, establishing both concept-specific features (such as unique visual differentiators, lighting, materials, and colors) and shared functional elements (including ventilation placement, fire hatches, transportation elements, and serviceability). To ensure alignment and rapid visualization, the team utilized real-time sketching alongside digital mood boards, referencing imagery, Pantone swatches, material examples, and official Volvo HEX color codes. To bring an example of how imagery was used, see inspiration collage on Figure 4 of how the team looked into different ways the system could display its operational status.



Figure 4: Status display inspiration collage

An overview of the workshop discussions and preliminary design agreements is provided below combined with final sketches for each concept. These served as the baseline for subsequent CAD modeling, prototyping, and rendering. A comprehensive breakdown of the finalized specifications and engineering metrics can be found in Section ??.

Sustainable Concept

The focus of the sustainability-inspired concept, was warm, eco-conscious design expressions that integrates well into surrounding environments, see Figure 5. The workshop discussions primarily revolved around the placement and structural viability of the wooden elements, exploring how to shape them while still accommodating standard corner castings. Material choices were debated between regional oak and MetaWood, a premium metal-based material with a wood-like finish (“MetaWood — Wood-Finish Metal Facade Cladding India”, n.d.). The team also brainstormed how to effectively conceal the required ventilation behind these exterior wood panels. Regarding lighting, the discussion focused on playing around with different color codes, lengths, and intensities for integrated LED strips. The initial layout exploration limited service access to the side door, while planning for battery rack assembly to take place through double doors with rounded corners on one of the long sides, though changes were later made to this setup.

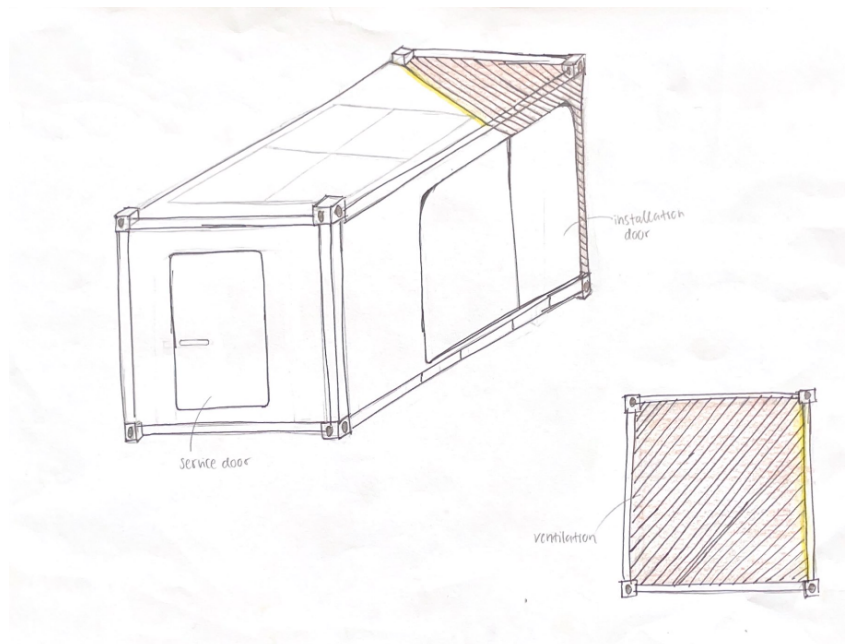


Figure 5: Final sketch of Sustainable

Industrial Concept

The Industrial Concept workshop discussion centered heavily on durability and robust, heavy-duty styling, see Figure 6. Brainstorming focused on a color-blocking strategy to visually separate the two functional zones of the BESS using a dual-tone industrial palette. To simplify manufacturing, the team evaluated the use of consistent chamfers and a traditional sharp-corner container silhouette instead of rounded profiles. For the ventilation aesthetics, the team compared various commercial Volvo truck grilles as direct design inspiration for the vent pattern. The structural layout discussions explored combining a standard long-side installation door with a heavy-duty industrial double-door on the short side for maintenance.

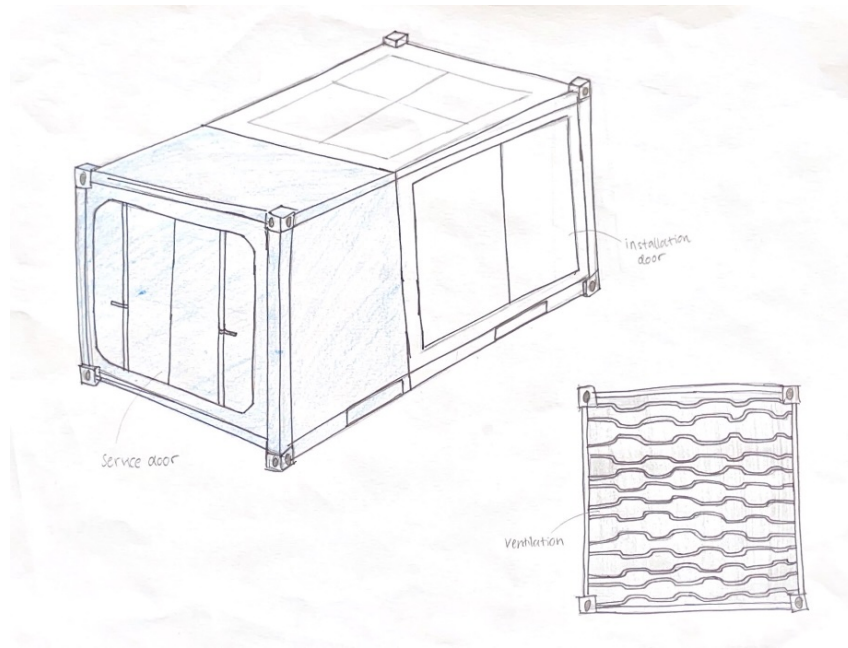


Figure 6: Final sketch of Industrial

Scandinavian Concept

Discussions for the Scandinavian concept focused heavily on extreme minimalism, clean lines, and creating a premium, seamless structural appearance, see Figure 7. To achieve this seamless look, the team brainstormed concealing the forklift pockets, like visible in Figure 8a, using spring-loaded covers inspired by restaurant kitchen doors as in Figure 8b, which would push open when used and snap back flush to the sides of the pockets instead of the top, therefore, not bearing any load when the BESS is lifted. However, this idea was ultimately abandoned during the session, as the team concluded that matching the colors and textures in production to ensure a good look would introduce unnecessary manufacturing complications without adding significant value. For the central piece on the short side, the team drew inspiration from soft-worn sea rocks and Volvo truck headlights, playing around with those forms to integrate a subtle, flush light feature. The initial access strategy explored a long-side door with rounded corners paired with a short-side double door, both of which were altered during the final modeling.

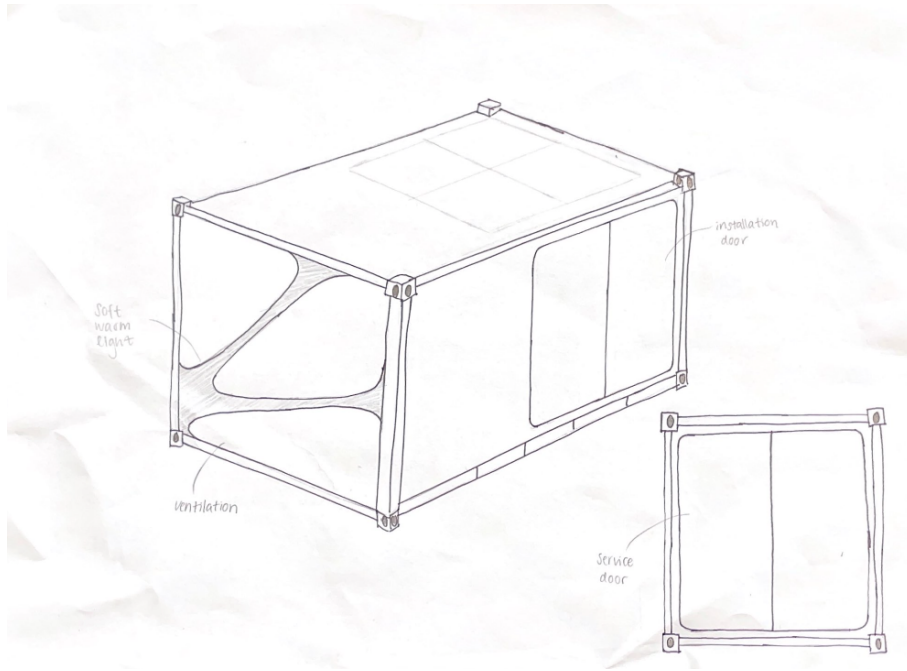


Figure 7: Final sketch of Scandinavian



(a) Forklift pocket covers



(b) Restaurant kitchen doors

Figure 8: Forklift covers and inspiration

5.2 Analysis of Alternative Solutions

To ensure a comprehensive exploration of the design space, while maintaining project momentum to make definitive decisions and move forward, the team conducted a continuous technical and aesthetic analysis of alternative solutions throughout the project's duration. This iterative evaluation occurred in parallel with the design's evolution. This agile approach allowed the team to constantly filter out non-viable ideas based on emerging constraints without stalling progression. This chapter documents the primary rationale behind these rapid eliminations and pivots, divided into two distinct tracks: the constant narrowing of the initial exterior design themes, followed by an overview of alternative manufacturing and structural architectures that were considered.

Exterior Aesthetic Directions and Thematic Evolution

When synthesizing initial sub-solutions into concrete concepts, random combinations were deliberately avoided. Instead, the team mapped targeted thematic directions, see Figure 9, ranging from "Functional" and "Realistic" to "Unexpected" and "Playful" - ensuring the generation of options that were both highly

creative and technically grounded. This exploration initially yielded 10 distinct concept directions, see Figure 10.



Figure 9: Thematic themes for concept generation

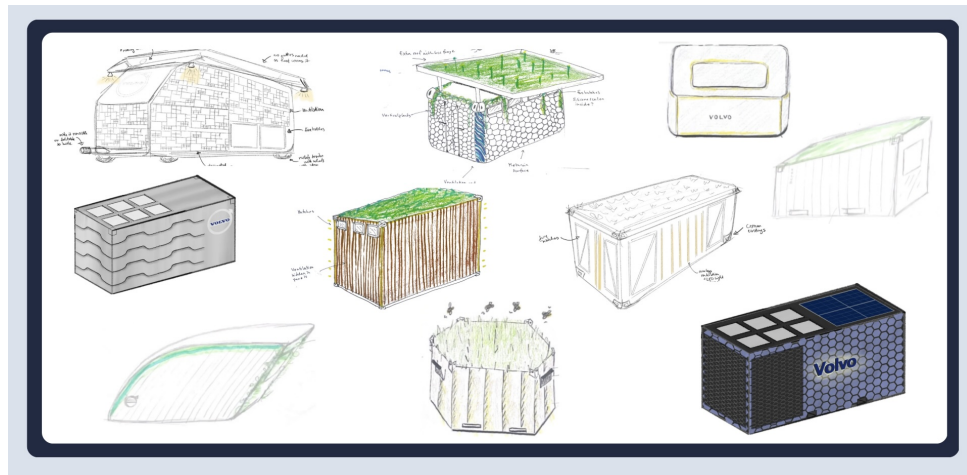


Figure 10: Initial concept direction ideas

A comprehensive group discussion was held to evaluate the 10 directions, focusing primarily on the viability of singular features and alignment with project goals. During this process, several concepts were eliminated due to their manufacturing or cost complexities. In addition, logistics and brand identity played a key role in decision making. Many configurations featured intricate geometries that showed to be too expensive or difficult to realize, which also compromised the core requirement of easy stacking and unit transport. Furthermore, these complex forms lacked a distinct "Volvo feel." Feedback from Volvo Energy reinforced these conclusions, emphasizing the need to simplify and tone down the overall design language. Ultimately, while these initial directions were eliminated, they served as critical stepping stones and the discussion- and evaluation of them resulted in increased knowledge and insights. Interestingly, many core elements of the final designs can already be observed at this stage, as they directly evolved from this initial collection of ideas.

Through this evaluation, the remaining ideas were clustered into four core themes: Sustainable, Sci-fi, Scandinavian, Industrial - to provide a clearer design trajectory. The team iterated on these four themes using reference imagery and focused sketching, see Figure 11. Ultimately, the "Sci-fi" concept was eliminated from the final selection. Although visually differentiated, it was deemed to an extent harsh and intimidating, directly contradicting the primary project aim of seamlessly integrating the unit into its surrounding environment. Furthermore, its futuristic aesthetic lacked the clean minimalism central to the Volvo design language. Consequently, this direction was dropped, leaving the three final concepts:

Sustainable, Industrial, and Scandinavian.

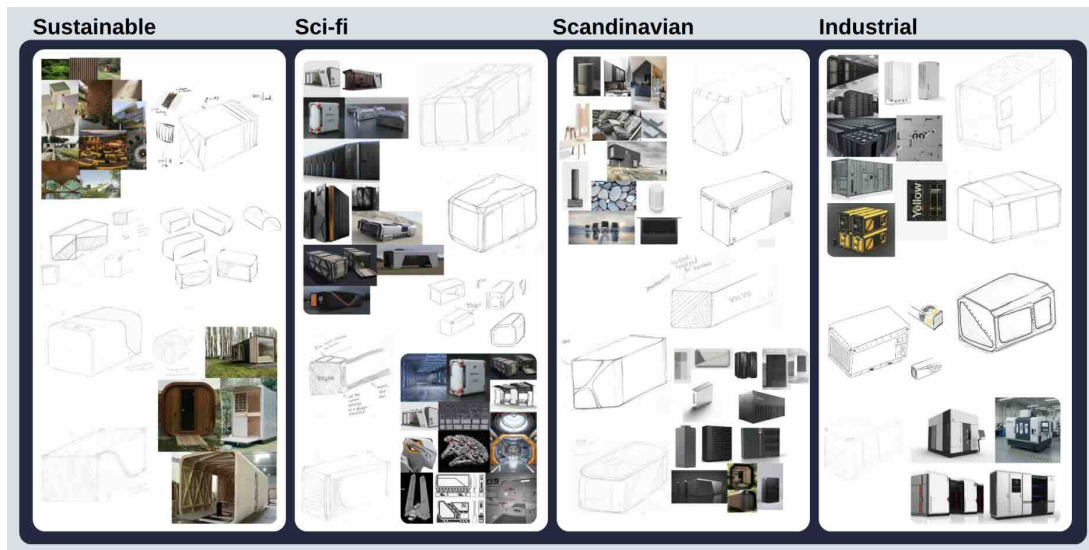


Figure 11: More focused concept directions

Alternative Manufacturing Principles and Modular Architectures

During the latter stages of the concept development phase, the team also explored innovations in modularity and manufacturing. Currently, Volvo Energy orders customized containers, that upon delivery go through interior assembly (batteries and electronics) into their container casings. To challenge this convention, the team had two modular ideas designed to streamline assembly and enhance structural flexibility, see Figure 12. Both ideas move away from the "fixed box" approach, instead utilizing a build-up method that prioritizes internal access during the manufacturing stage.

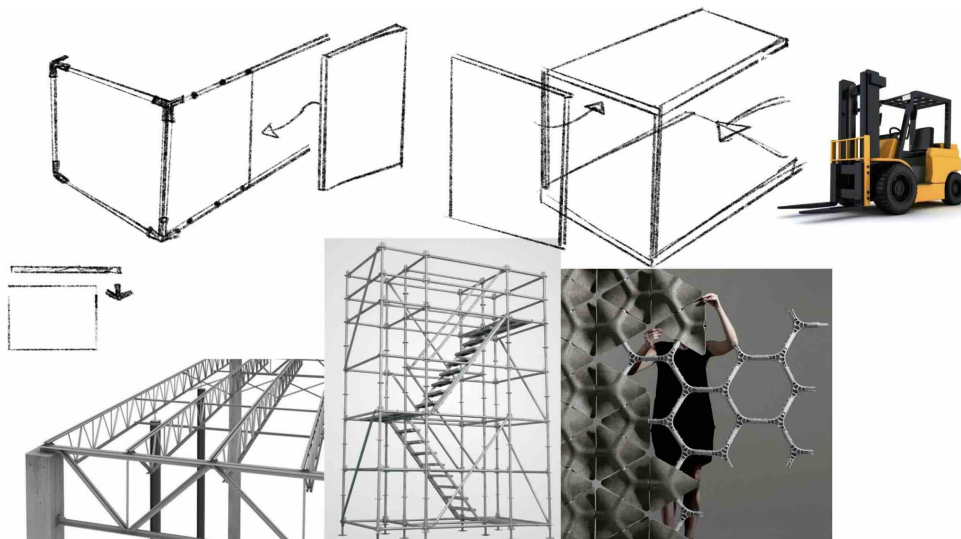


Figure 12: Truss-frame and base-and-pillar manufacturing concepts

Firstly, a truss-frame system inspired by scaffolding and truss architectures, utilizing a general skeletal frame. The assembly sequence begins with the construction of the frame, followed by attachment of the base plate and then the installation of all internal components. Once the interior is finished, the exterior is sealed using modular plates secured by bolts or other high-strength fasteners. This system is expandable, allowing for easy scaling. The second one uses a base-and-pillar idea and is considered the more viable of the two. The design utilizes a reinforced base as the foundation, where vertical side poles (pillars) are attached to this base, then the interior components are assembled onto the base, before the

exterior plates are fitted onto pillars. Again, this provides maximum clearance from all sides for assembly teams and equipment.

The primary driver for these modular designs was the optimization of the assembly process. Currently, the use of pre-built containers necessitates the use of heavy forklifts to maneuver batteries through narrow doors and calls for a need for large, industrial-style access doors that detract from the unit's aesthetics. Thus key benefits of the modular approach would be, firstly, unobstructed access, by installing side plates after the heavy internal components are set and eliminating the need for specialized "ugly" loading doors. Secondly, potential aesthetic flexibility as the exterior plates can be customized with different materials or finishes to create various design variants, resulting in a more polished, premium service opportunity for customers. Lastly, it would be viable as the battery and service rooms only require access approximately once a year, the removable plate system could provide sufficient accessibility for scheduled maintenance.

These ideas were pitched to Volvo Energy, where they received positive feedback as they presented an innovative approach to scalability and manufacturing efficiency. However, due to the project's strict time constraints and priorities on aesthetics, the decision was made to focus on the original, established manufacturing method for the current iteration. While the team reverted to the traditional container-based method for the remainder of the project, the modular insights gained remain highly valuable as possible future innovations for product architecture, offering opportunities toward more efficient- and aesthetically versatile energy storage solutions.

5.3 Technical Problems and Verifications of Technical Systems

As the three concepts evolved into feasible design solutions, the team encountered several technical bottlenecks inherent to large-scale infrastructure projects of this nature.

A primary technical challenge emerged from treating the ventilation layout as a core design element rather than a purely functional. Battery Energy Storage Systems generate significant heat during operation, demanding high-airflow ventilation, yet because these units are deployed outdoors, the enclosure must remain fully sealed against harsh environmental elements like rain, dust, and snow. For the Sustainable and Scandinavian concepts, concealing the ventilation to achieve aesthetic seamlessness significantly complicated airflow optimization, while the Industrial concept moved away from classical louvers toward a custom-patterned grille. Ensuring that these stylized geometries remain fully functional without compromising cooling performance requires precise alignment. For instance, the Sustainable concept's wooden cover utilizes a layered geometry to create a labyrinth effect, allowing air to flow freely while physically blocking water ingress.

This challenge is further complicated by layout changes. In previous configurations, the BESS architecture distributed ventilation across three different sides. In contrast, the new concepts consolidate the ventilation entirely onto a single side. This structural shift alters internal air currents and necessitates airflow testing to verify that the new centralized layout and custom vent patterns provide sufficient flow. Furthermore, thermal management considerations directly restricted the color palette selection. While the team initially explored a wide variety of colors, darker ones absorb substantial solar radiation, compromising internal temperature control. To maximize thermal reflection, the primary body colors across all concepts were strictly kept light. The only exception is the dark blue accent on the Industrial concept; however, this dark panel is functionally restricted to the service room rather than the battery racks, mitigating thermal risks to the sensitive battery cells.

Beyond thermal management, the introduction natural wood in the system presented an external fire risk. To mitigate this danger of the unit catching fire from external sources, the team evaluated MetaWood, a premium metal substrate with an authentic wood finish ("MetaWood — Wood-Finish Metal Facade Cladding India", n.d.). Using this material as a substitution for wood would effectively eliminate the external fire hazard while preserving the warm visual expression that the wood brings to the Sustainable concept.

Crucially, because MetaWood is fundamentally a metal-based solution, it would also eliminate the manufacturing- and structural risks associated with differential thermal expansion. Had natural timber been rigidly welded or bolted directly to the main structural steel, the expansion- and contraction rates during temperature fluctuations would cause the outer panels to warp or the fasteners to shear. By utilizing MetaWood, these thermal expansion rates are inherently matched to the steel frame.

Finally, another aspect the team had to carefully consider was the long-term environmental durability of

the exterior finishes. Fortunately, it was concluded that a typical BESS unit has an operational lifespan of approximately 15 years, which simplifies the requirements, as the exterior does not need to be engineered to exceed this time frame.

5.4 Final Concepts

In total, three concepts with different themes were developed in order to explore different directions and fully utilize the design space: Industrial inspired, Scandinavian inspired, and Sustainable inspired. All three designs used the same base as the original PU2000, which includes the base plate and frame with corner castings and forklift pockets. Additionally, all three concepts share a few common design elements, including the fire hatches, long side doors, and a short side completely dedicated to ventilation, but with different designs. On the top, each design had four fire hatches placed in the same square pattern. On one of the long sides, there was a double door wide enough to place the battery packs that is to be bolted shut after installation. The other long side was left clean to allow either customer or Volvo branding. Each concept is explained in detail below.

5.4.1 Industrial Inspired

The first concept developed was the Industrial, intended to capture the classic feeling often found within industrial contexts but with a cleaner and more modern style. Main inspiration and aesthetic design elements were taken from Dyson and their way of showcasing the internal functions of their products. In this case, this was done through color blocking with a dark blue color (HEX: # 202A44, dividing the exterior into one part for the batteries and one part for the service and maintenance room. The base color chosen for this concept was a light gray color (HEX: # E1DFDD). On the short side dedicated to the service room, the original shipping container doors were kept as a way of embracing the industrial theme. The other short side, dedicated to ventilation, was designed with a pattern imitating the one found on the grill of Volvo Trucks. This pattern covered the ventilation, with the actual panels being slightly visible between the gaps in the pattern. The final result of the 3D-model for this concept is illustrated in Figure 13.



Figure 13: Renderings of the industrial-inspired BESS CAD model, displaying exterior angles.

The main material of this concept was the standard steel currently used for the PU2000 called Corten steel, which is the standard material used to manufacture 20-foot shipping containers (“What Are Shipping Containers Made Of? Container Materials”, n.d.). Corten steel is a low-alloy steel, well known for its great resistance toward electrochemical corrosion, making it highly weather resistant (“Corten Steel: Everything You Need to Know About This Material — Delmet SRL”, n.d.). The most significant difference in the material, for all concepts, were the color and the surface. Standard shipping containers have a corrugated surface, whereas all the concepts used a flat surface for easy maintenance. The ventilation also used standard materials and ventilation panels currently in use on the existing PU2000, and for the grill

pattern, stainless steel was chosen. During the project, the material used for the 3D models and general estimations has been low alloy steel (40CrMoV13-9), which was identified using Granta Edupack. This material was used as it had similar properties to the actual material used, Corten steel. The complete data sheet for low alloy steel 40CrMoV13-9 can be found in Appendix B.

As previously mentioned, this concept aimed to capture a modernized and cleaner industrial feel. This makes it highly suitable for more utilitarian and industrial environments, where it can provide a reliable source of energy. The clean and straightforward design easily blends in with other elements often found at rest stops along highways. Therefore, the Industrial inspired concept was rendered at a charging station located next to a highway. In Figure 14a, the focal point is a rest stop with an electrical charging station for both passenger cars as well as larger trucks. This type of user environment has a lower level of interaction with the public. Users simply stop to charge while traveling long distances along the highways. Therefore, the design would most likely blend in with the surrounding environment and not be as noticeable to passersby. In Figure 14b, the whole product family is rendered to demonstrate how the design can be translated across multiple different sizes to ensure scalability and versatility. The different sizes are the standard 20-foot container, a 10-foot container, and a smaller modular unit. This showcases the design's ability to adapt while still keeping the general design language.



(a) Industrial-inspired render of the BESS concept in a suitable environment. The BESS unit is a custom CAD model, the surrounding environment was generated by AI (Google, 2026a).



(b) Industrial-inspired render of the BESS product family. The central unit is a custom CAD model, while the two variations and the environment were generated by AI (Google, 2026b).

Figure 14: Renders of industrial-inspired BESS concept.

5.4.2 Scandinavian Inspired

The second concept developed was the Scandinavian, inspired by the bright, crisp, and soft elements of Scandinavian nature and design. This concept aimed to capture a minimalist and sleek design language that easily blends into its surroundings without dominating the space. To achieve this, the same light gray color used in the Industrial inspired design was chosen for this design as well. The difference was that the entire design was the same color, which further provides a minimalist and low tone aesthetic.

The focal point of this design was the short side dedicated to ventilation with integrated LED-lighting, which was inspired by sea-worn stones and the headlight of a Volvo truck. The ventilation itself would be partially hidden behind the stone details, with some of the vents visible, and the LED-lighting creates a warm and cozy ambiance. The final result of the 3D-model for this concept is illustrated in Figure 15.



Figure 15: Rendering of the Scandinavian-inspired BESS CAD model, displaying exterior angles.

The main material used for this concept was the same stainless steel used in the Industrial inspired design. This concept used the same material for the ventilation located behind the rock design as well. For the integrated LED-strips, a similar solution used for seamlessly integrating ambient lighting in furniture and artwork was chosen. Electrical wiring, cables, and other components would be directed on the backside of the rock design to be invisible from the outside.

The aim of the Scandinavian inspired concept was, as previously mentioned, to create a minimalist and sleek design that easily blends in with its surrounding environment. This makes it suitable for a wide variety of user environments with different levels of public interaction. The renders were created in an environment where the focal point is a charging station close to the city core of Gothenburg. In Figure 16a, there are a few smaller charging stations intended for passenger cars and a charging station for trucks. In the background, there is a small hub where users can wait while charging. In order to showcase the ambient LED-lighting, the setting imitates early evening, resulting in a warm and cozy light emitting from the ventilation side. In this kind of environment, public interaction may differ between low and high levels depending on multiple factors, such as the time of day, time of year, and location in relation to the city. For example, if the environment were located close to a park, the level of public interaction would most likely be very high due to people frequently walking by. This would, in turn, make the public perception and message conveyed by the design highly important and could have significant impact on the public perception of the company providing the BESS. However, if the environment were located in a less frequented area, the level of public interaction would be lower. In turn, this would make the impact of public perception slightly lower. The minimalist and quiet design makes it blend in well with the surrounding environment, while the LED-lights remind passersby that it is there. In Figure 16b, the complete product family is rendered in the same environment. The different sizes are the same as for the Industrial inspired concept.



(a) Scandinavian-inspired render of the BESS concept in a suitable environment. The BESS unit is a custom CAD model, the surrounding environment was generated by AI (Google, 2026c).



(b) Scandinavian-inspired render of the BESS product family. The central unit is a custom CAD model, while the two variations and the environment were generated by AI (Google, 2026d)

Figure 16: Renders of Scandinavian-inspired BESS concept.

5.4.3 Sustainable Inspired

The third concept developed was the Sustainable inspired concept, which aimed to visually communicate sustainability, warmth, and environmental integration. This concept was heavily inspired by the colors and materials of nature, in combination with design elements from the World of Volvo. The base color for this concept was a cream white color (HEX: # F0EEE9). The focal point of this concept is the wooden panel with integrated LED-lighting that covers the short side dedicated to ventilation, part of the roof, and part of one long side. The ventilation would be completely hidden behind the wood panels, with the pattern of the panels being continuous on the roof and long side. The LED-lighting creates a similar warm and cozy ambiance as mentioned for the Scandinavian inspired design, the only difference being the placement of the LED-strips. They would be placed along the front and top edge of the wooden panels to spread light on the front of the BESS. The final result of the 3D-model for this concept is illustrated in Figure 17.



Figure 17: Renderings of the sustainable-inspired BESS CAD model, displaying exterior angles.

The main material used for this concept was the same stainless steel that was used for the previous two concepts, but with a cream white base color instead of light gray. This concept also used the same material for the ventilation as the previous two concepts. For the wooden panels, there were mainly two different options available, each with various advantages and disadvantages. One option was to use a similar wooden material that LexEnergy, a Swedish-based company, uses for their product LexHub (“Home - LexEnergy - The future of energy solutions”, n.d.). Their design uses a hardwood panel for their charging stations, indicating good durability that is compatible with the Swedish weather and climate. However, real wood will deteriorate and become worn over time, likely becoming scratched and damaged depending on where it is installed. Additionally, using real wood could potentially create a fire hazard and make it more susceptible to vandalism. It is important to note that the wooden panels would not be attached prior to transportation and installation on site, but would be mounted last. Another option was to use a material manufactured by a company called Metaguisse (“MetaWood — Wood-Finish Metal Facade Cladding India”, n.d.). The material was called MetaWood, which is a metal based wood finish, creating the warm and inviting look of wood but with the durability of metal. MetaWood is weather-proof and fire-resistant, which indicates longer durability with less deterioration over time compared to real wood.

As previously mentioned, the aim of the Sustainable inspired concept was to visually communicate sustainability, warmth, and environmental integration. This makes this design suitable for a wide variety of user environments, ranging from lively urban areas to slower paced suburban areas with different levels of public interaction. The renders for this design were created in an environment where the focal point is a smaller parking lot designed for passenger cars, located in a suburban area. In Figure 18a, there are multiple smaller charging stations with cars connected and charging. There are a few modern-looking suburban buildings in the background, creating a quiet and calm environment. Similar to the renders for the Scandinavian inspired concept, the setting imitates early evening to showcase the LED-lighting integrated with the oak panel. This creates a warm and ambient light emanating from the front and top of the BESS. The level of public interaction in this environment would be relatively high, since users charging their cars may frequently use this particular charging station. In turn, this makes the public perception and message conveyed by the design highly important. Additionally, since the area is suburban, it is likely that other people frequent the area as well, increasing the level of public interaction. The warm and inviting design flows well with its surrounding environment, while the oak panel and LED-lighting gently brings out its presence. In Figure 18b, the complete product family is rendered in a similar environment imitating a suburban residential area. The different sizes are the same used for the previous two concepts.



(a) Sustainable-inspired render of the BESS concept in a suitable environment. The BESS unit is a custom CAD model, the surrounding environment was generated by AI (Google, 2026e).



(b) Sustainable-inspired render of the BESS product family. The central unit is a custom CAD model, while the two variations and the environment were generated by AI (Google, 2026f).

Figure 18: Renders of sustainable-inspired BESS concept.

5.5 Patent Analysis

This section analyzes the patent landscape for the BESS exterior to identify technical limitations and evaluate the possibilities for future design protection for the final design solutions.

5.5.1 Patent Landscape Analysis

In a previous study, a targeted patent analysis was conducted that focused exclusively on utility patent for critical mechanical interfaces: *Pressure Relief*, *Outdoor Ventilation*, and *Physical Security* (Arvidsson et al., 2025). The analysis showed that complex mechanical solutions, such as pressure relief and advanced ventilation, are largely limited by existing intellectual property rights (Arvidsson et al., 2025). The conclusion from this investigation was that these technical components should be purchased as ready-made standard solutions to avoid patent infringement. Physical security was assessed to be a mature technology without direct limitations. However, it is important to note that no design patents or competitors' visual patents were examined.

5.5.2 The Role of Patents in Concept Development

In the development of the final concepts, previous patent analysis has not been a guiding factor in the design. Instead of focusing on or being limited by moving mechanical details, the design work was based on the overall visual impression of the exterior with mechanical solutions in mind.

5.5.3 Potential for Design Patents

Since no new mechanical or functional solutions have been constructed, functional patents are not relevant to the developed concepts. However, there is a possibility of securing design patents for the final result.

5.6 Manufacturing Methods and Adaptability

This section describes how the design concepts are adapted to existing production and logistics chains, the choice of manufacturing methods, and the added value that the design contributes to the overall offering.

5.6.1 Adaptation to Production and Logistics

A key requirement for the design concepts has been adaptability to existing production and logistics chains. The production process is based on a modified 20-foot container being manufactured and ordered according to the specific drawings. This container is then sent to Volvo Energy’s facility in Skövde, where all internal equipment is installed. In Skövde, the additional exterior components that have been designed to cover the ventilation and create the unique visual expression are also assembled. Since this process largely reflects how Volvo Energy’s existing units are manufactured today, the new concepts involve minimal changes to the production line.

5.6.2 Manufacturing Methods and Material Selection

When it comes to manufacturing methods, the concepts offer some flexibility. The added exterior components that cover the ventilation can either be manufactured in-house or purchased from subcontractors, and then finally assembled on the container in Skövde. To realize the production of the new design, three things are primarily required: the creation of new modified construction drawings that are sent to the container supplier, the procurement or manufacture of the specific exterior panels, and the introduction of a final assembly step in Skövde, where the panels are mounted on the container.

In terms of materials, corten steel, which is a low alloy steel (40CrMoV13-9), is the primary choice for the basic structure and panels to guaranty robustness, stability, and durability. Corten steel is the most common material used for containers (“What Are Shipping Containers Made Of? Container Materials”, n.d.). For the Sustainable inspired concept with a ribbed facade, there are several material options; for example, real wood can be used for a softer expression, but to meet stricter fire requirements, fire-resistant metal with wood imitations is also considered (“MetaWood — Wood-Finish Metal Facade Cladding India”, n.d.).

5.7 Complete Requirements Specification

This section presents the complete requirements specification for the project. The specification is largely based on insights and customer needs identified through stakeholder interviews in a previous study by Arvidsson et al. (2025). This collected data has subsequently served as a framework for the development of the BESS unit exterior. As new perspectives and a deeper understanding have emerged in the project, the requirements have been continuously evaluated and updated during the work to constantly steer towards the most optimal end result possible.

The requirements specification table, illustrated in Figure 19 and 20, presents a comprehensive summary of all technical, aesthetic and functional requirements as well as identified customer desires. This summary has been a central tool in the work of balancing Volvo Energy’s brand identity with the industry’s strict safety regulations and the need for smooth environmental integration. By clearly mapping out what are absolute requirements versus value-added add-ons, the design process has been consistently steered towards concepts that are both innovative and technically feasible.

Criterion		Goal value	R/D	Weight (1-5)	Customer needs addressed	Method of verification
	Direction					
1. Performance						
1.1 Internal temperature	Maintain the internal temperature of batteries for an optimal performance	23 degrees celsius	D	5	Internal temperature regulation for optimal performance. (MB)	Testing - Simulations
2. Product environment						
2.1 Temperature regulation	Ability to handle outdoor temperature span	-20 - +40 degrees celsius	D	5	Internal temperature regulation for optimal performance. (MB) Internal temperature regulation based on external temperature. (MB)	Testing - Simulations
2.2 Humid and wet environments	Withstand high humidity and protect internal components from moisture caused by rain and snow	IP55	D	5	Prevent water leaks from high humidity, rain and snow. (MB)	Testing
2.3 Dust and polluted environment	Prevent dust and pollution from damaging critical components	IP55	D	5	Prevent dust and pollution from damaging critical components. (MB)	Testing
3. Lifespan						
3.1 Minimum life span	Have a minimum lifespan of 15 years	15 years	R		An environmentally sustainable exterior construction and design. (OD) An enduring interior and exterior design. (OD) An enduring interior and exterior design. (OD) A design that mediates a sense of quality, stability, and robustness. (A)	LCA
4. Size						
4.1 Exterior dimension	Measurements within the dimensions of a standardized 20ft container	6.06x2.44x2.59 [m]	R			Control
5. Safety						
5.1 Fulfill required fire safety laws	Fulfill all required fire safety laws		R		Prevent fire spread and ensure a controlled sequence of events. (MB)	Testing
5.2 Fulfill CSC 8	Fulfill CSC 8 regulations		R			Testing
5.3 Prevent bodily injury	Ensure safe service, usage and interaction without bodily injuries		R		Prevent bodily injury on: service personnel, end users, and passersby. (MB)	Control
6. Aesthetics						
6.1 Design mediates positive associations	A design that mediates positive and reassuring associations		D	3	A design that mediates positive and reassuring associations (A)	Testing - User interviews, showing prototypes
6.2 Design mediating quality, stability and robustness	A design that mediates quality, stability, and robustness		D	3	A design that mediates a sense of quality, stability, and robustness (A)	Testing - User interviews, showing prototypes
6.3 Intuitive exterior design	An intuitive exterior design		D	2	An intuitive exterior design. (A) A design that allows modification for smooth integration with nearby structures. (A)	Testing - Surveys, user interviews - Qualitative evaluation
6.4 Modern and high-tech design	A design that mediates an up-to-date expression with high-tech design language		D	3	A design that mediates positive and reassuring association. (A)	Testing - Surveys, user interviews - Qualitative evaluation
6.5 Branding and marketing	An exterior design that allows branding and marketing		D	4	An exterior surface that allows branding and marketing. (A) An exterior that facilitates cleaning. (A)	Testing
6.6 Modular design	A design that allows modification for smooth integration with nearby structures		D	2	Service modularity. (OD) Mobility, and availability to move the BESS unit on-site. (OD) Simple installation process. (OD) A design that allows modification for smooth integration with nearby structures. (A)	Testing
6.7 Display information	An exterior that can display information (LED-screen, QR-code etc.)		D	2	An exterior design with a modern and high-tech feel. (A) An exterior surface that allows branding and marketing. (A)	Testing

Figure 19: Requirements specification

Criterion		Goal value	R/D	Weight (1-5)	Customer needs addressed	Method of verification
7. Materials						
7.1 Exterior that facilitates temperature regulation.	An exterior designed to facilitate temperature regulation by reflecting heat and aid in maintaining optimal performance		D	5	An exterior that reflects sunlight, in order to maintain a controlled temperature. (OD) An environmentally sustainable exterior construction and design. (OD) An environmentally sustainable choice in materials. (OD)	Testing
8. Quality						
8.1 Scratch resistant exterior surface	Make it difficult to create scratches on the surface		D	2	An exterior that reflects sunlight, in order to maintain a controlled temperature. (OD)	Testing
8.2 Production cost						
8.3 Manufacturing cost	Maximum cost for manufacturing of the exterior	300 000 SEK	D	5		Estimated calculation
9. Maintenance						
9.1 Service modularity	Ability to change individual modules and components		D	3	Service modularity. (OD) A design that allows modification for smooth integration with nearby structures. (A) Separate BESS unit and control room. (A)	Control
9.2 Service availability	Ability to reach all service components, without affecting performance or safety		D	5	Service modularity. (OD) A design that allows modification for smooth integration with nearby structures. (A) Separate BESS unit and control room. (A)	Control
9.3 Installation availability	Minimal installation steps required by the customer		D	3	Simple installation process. (OD)	Control
9.4 Facilitates cleaning	An exterior design that facilitates cleaning		D	3	An exterior that facilitates cleaning. (A)	Testing
10. Competitor solutions						
10.1 Differentiate from competitors in terms of exterior design	Move away from the common design of current competitor solutions (white colour, container look, etc)		D	5		Testing - Surveys, Customer interviews - Qualitative evaluation
11. Transportation						
11.1 Freight	Ability to be transported from factory by standard freight	Minimum 1 time	R			Control
11.2 Mobility	Ability to move around after initial installation		D	2	Service availability. (OD) Service modularity. (OD) Mobility, and availability to move the BESS unit on-site. (OD)	Control
12. Sustainability						
12.1 Material	Minimize environmental impact of materials		D	3	An environmentally sustainable choice in materials. (OD) An environmentally sustainable exterior construction and design. (OD) A design that mediates positive and reassuring associations. (OD)	LCA
12.2 Construction and design	Construction and design made to minimize environmental impact		D	3	An environmentally sustainable choice in materials. (OD) An environmentally sustainable exterior construction and design. (OD) A design that mediates positive and reassuring associations. (OD)	LCA

Figure 20: Requirements specification

5.7.1 Customer Demands and Wishes

Customer needs, regarding the exterior of the BESS unit, have been identified during a previous study through interviews with stakeholders (Arvidsson et al., 2025). Since the end customer primarily consists of other companies in the charging infrastructure and commercial and industrial segments, the product must balance strict industrial functional requirements with strategic and aesthetic wishes.

The needs was categorized according to the KANO-model into basic requirements (must be), performance requirements (One-Dimensional), and attractive added values (Attractive) (Sauerwein et al., 1996). The compiled list of the identified customer needs is presented in Table 1 below.

Table 1: Customer needs list (Arvidsson et al., 2025)

Must-be	One-dimensional	Attractive
Internal temperature regulation for optimal performance	An exterior that reflects sunlight, in order to maintain a controlled temperature	A design that mediates positive and reassuring associations
Internal temperature regulation based on external temperature	A scratch resistant exterior surface	A design that mediates a sense of quality, stability, and robustness
Prevent water leaks from high humidity, rain and snow	Service modularity	An intuitive exterior design
Prevent bodily injury on: service personnel, end users, and passerbys	Service availability	An exterior design with a modern and high-tech
Prevent fire spread and ensure a controlled sequence of events	Simple installation process	An exterior surface that allows branding and marketing
Prevent dust and pollution from damaging critical components	An enduring interior and exterior design	A design that allows modification for smooth integration with nearby structures
	Mobility, and availability to move the BESS unit on-site	An exterior that can display information (LED-screen, QR-code etc.)
	An environmentally sustainable exterior construction and design	Separate BESS unit and control room
	An environmentally sustainable choice in materials	An exterior that facilitates cleaning

Must-be and **One-dimensional** focus largely on protection, safety, and operational reliability, which are the foundation for the existence of the BESS unit. The main task of the exterior is to protect the internal components by preventing the ingress of water, snow, dust, and contaminants. Safety is a non-negotiable requirement; the exterior must prevent bodily injury to both service personnel and the public, be designed to prevent the spread of fire, and ensure a controlled course of events in the event of a fire. Furthermore, customers demand durability, such as scratch-resistant surfaces and a design that lasts over time, while the unit must maintain its transportability and be easy to lift, install, and move when needed.

Wishes are an addition to the basic protective functions, the analysis shows a strong need for "soft" values and attractive values. Corporate customers want an exterior that conveys a sense of quality, stability, and robustness, while at the same time feeling modern and high-tech. One of the strongest wishes is that the design should be adaptable to blend smoothly into nearby surroundings and structures. In addition, there is a clear desire from customers to use the BESS unit's surface for their own branding and marketing.

5.7.2 Technical Requirements

A BESS is a deeply complex product whose internal components, such as battery chemistry, power electronics, software and cabling, are regulated by an extensive network of international standards and directives. However, for this project, the focus is on the exterior of the unit and its interaction with the environment. For this reason, the technical requirements have been narrowed down to include the specific standards that directly dictate the exterior design, safety features, weather protection and transportability.

To ensure that the exterior design is feasible and meets global regulatory requirements, the concepts have been developed while taking into account the classification and regulations presented in Table 2.

Table 2: Compilation of technical standards and regulation relevant to the BESS unit exterior and overall system integration.

Standard	Short description
UL 1642	Covers safety of lithium cells, including electrical, mechanical, and environmental tests.
UN 38.3	Transport regulation ensuring lithium cells and batteries can withstand thermal, mechanical, and electrical stresses.
IEC 60529	Defines IP (Ingress Protection) ratings for dust and water resistance.
UN 3480	Regulates transport labeling and packaging for lithium-ion batteries shipped alone.
IEC 62485-5	Safety for battery installations including ventilation, layout, and electrical hazards.
IEC 62933-2-1/ TS 62933-5-1	Safety, performance, and testing guidelines for energy storage systems (ESS).
UL 9540	Certification for energy storage systems covering overall safety integration.
NFPA 855	Fire and installation code for stationary energy storage systems.
NFPA 69	Explosion prevention systems for combustible environments.
NFPA 68	Venting guidance for deflagration pressure relief.
UN 1364/ UN 3536	Transport and hazard labeling for large battery systems and equipment.
UL 9540A	Test method for evaluating thermal runaway and fire propagation behavior.

5.7.3 Sustainability Requirements

Unlike the strict technical standards for battery safety and transportation, the sustainability work for the BESS unit exterior has not been primarily dictated by mandatory legal requirements. Instead, the project has been based on actively pursuing a design that aligns with Volvo Energy’s internal guidelines and overall sustainability goals (Volvo Group, 2025). As sustainability is deeply integrated into Volvo Energy’s business model, the development of the design concepts has been carried out with a continuous effort to balance the classic sustainable triangle: environmental, social, and economic aspects.

Environmental: The design should prioritize the use of recyclable materials and ensure exceptionally robust construction. The exterior needs to withstand the wear and tear expected during the product’s lifetime to minimize the need for resource-intensive repairs.

Societal: The product must facilitate the green transition without causing disruption to the local environment. A key requirement is to create an aesthetically pleasing exterior that lowers the general acceptance threshold for placing units in urban or public environments. The design must harmonize with its surroundings while maintaining uncompromising safety for the people moving around the unit.

Economic: For the product to drive sustainability at scale, the design must be commercially and economically viable. This requires integration with existing production methods and supply chains to eliminate the need for expensive custom solutions or factory conversions. Furthermore, material optimization and standardization are required to keep production and maintenance cost down for the end customer.

5.7.4 Ethical Considerations

Energy storage systems like BESS play a pivotal role in the ongoing green transition, but the rapid market growth brings with it a number of difficult ethical dilemmas. Just as previous studies have identified (Arvidsson et al., 2025), technological development must go hand in hand with much greater responsibility. This includes ensuring sustainable and responsible supply chains, guaranteeing physical security, and working towards a natural integration into society.

Perhaps the greatest moral challenge for the industry is the clash between doing good for the climate globally and protecting human rights locally. Right now, lithium-ion batteries are a very important part of our transition to renewable energy, but the path to that end often has dark sides. Those working in extraction and manufacturing are exposed to dangerous emissions and health risks. In addition, in several

production countries, child labor, forced labor, and wages that are barely livable occur (Hussain, 2025). This involves a difficult balancing act, where the technology that will drive climate change forward also risks causing harm to vulnerable populations. This means that manufacturers really have to step up and take much clearer ethical responsibility.

At the same time, the market is starting to react to these shortcomings. The demand for traceability and responsibly sourced batteries is now growing rapidly, largely due to stricter legislation, but also thanks to an increasingly aware group of consumers and stakeholders (Patsnap, 2025).

Another important ethical responsibility concerns physical security. Placing large-scale energy storage in society clearly weighs heavily on the utility of the electricity grid, but this must be balanced against the potential risks to the public. Challenges around fires and thermal runaway mean that extremely reliable security measures are not just a technical requirement, but a moral imperative (Montel, 2024). From this perspective, the exterior has a much more important task than just being aesthetically pleasing. The design must be structured in a way that ethically and practically guarantees the protection of all those moving in the vicinity.

5.8 Survey

To obtain a better understanding of the public perception of each design, a survey focusing on the message conveyed by the design was conducted. The purpose was to quantitatively gather input to assess the aesthetic and visual perception of the concepts. The target group for the survey was passersby who would likely engage in incidental and peripheral interaction, rather than purchasing customers. Given the previously mentioned increasing electrification, the BESS is likely to be integrated into public environments and areas in the future. This integration will increase their visual impact on the general public, making community opinions more critical to long-term acceptance than those of the immediate purchasing buyers. The survey consisted of nine Likert-scale questions ranging from *Strongly Disagree* to *Strongly Agree*, alongside the option of *I don't know* to avoid forced responses. The questions included statements assessing the emotions and messages projected by the designs. To ensure construct validity and avoid miscommunication, key terms were explicitly defined within each question. An optional open-ended text field was also included for qualitative feedback. The survey is linked in Appendix C. The survey was distributed both digitally via a QR code and physically through posters during the final exhibition. In total, ten respondents answered the survey.

The result of the survey presented in detail is illustrated in D. Based on the answers from the survey, some conclusions regarding each concept could be drawn. The overall perception of the Industrial inspired concept was a durable secure design with a positive perception. The strengths of the Industrial inspired concept were safety and longevity. A majority agrees that the concept felt **Stable**, **Safe**, and **Longevity**, as demonstrated in Figure 21 below.

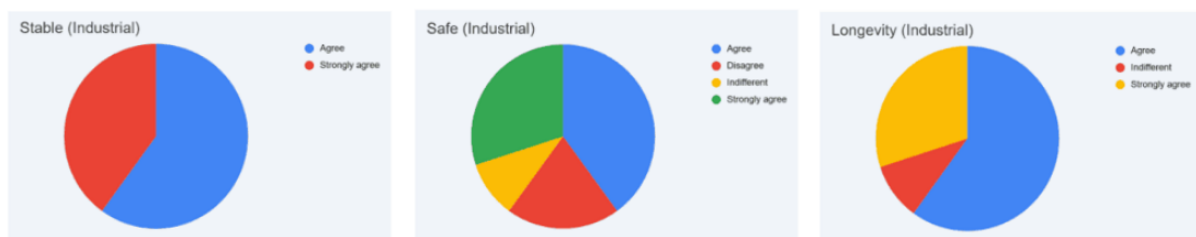


Figure 21: Survey result, Industrial inspired concept

Furthermore, a majority agreed that the concept added a higher value to the overall product. This indicated that most of the respondents viewed the concept as structurally viable. However, some weaknesses of this concept were the emotional response to the design. A majority disagreed that the design was **Calm** or **Subtle**, which indicates that the aesthetic appeal may be lower than expected. The overall perception of the Scandinavian inspired concept was a powerful but also polarizing design. The strengths of the Scandinavian inspired concept were its status and aesthetic appeal. A majority agreed that the design was **Safe** and **Subtle**, indicating a smoothly integrated design. Furthermore, a majority agreed that the concept expressed **Longevity** and added a higher value to the overall product. However, some

weaknesses of this concept majority were the perceived stability and aesthetic perception. While a majority agreed that the concept was stable, a polarizing opinion strongly disagreed with the statement. Indicating that this design may be a deal breaker. The overall perception of the Sustainable inspired concept was a well balanced concept with an exceptionally positive perception. The strengths of the Sustainable inspired concept were the emotional and durable perception. A majority agreed that the design was **Stable**, expressed **Confidence**, and **Calm**. This indicates an emotional resonance with the design, aligning well with the aesthetic appeal intended. However, some weaknesses of this concept were the perceived longevity and subtleness of the concept. While a majority agreed the design expressed **Longevity** and was **Subtle**, a few opinions disagreed. This indicates that the design may be perceived as less durable and loud.

5.9 Justification of Final Concepts

The final concepts are visualized in Figure 22 below.



Figure 22: Final concepts

Due to the project scope being confined to the exterior design, the internal architecture, internal material composition, and performance-related components were left unaltered in all the final concepts. The interior layout, design, and components is the same as found in the PU2000, which already fulfills the requirements. Therefore, by assuming the internal structure remains compatible with the new design suggestions, all technical requirements defined in Table 2 remain fulfilled. Additionally, all requirements related to safety labeling can also be assumed to be fulfilled as they can simply be attached to the exterior surface. However, since the labeling was not included in any of the prototypes, there is uncertainty about how this labeling would affect the overall perception of the designs.

Furthermore, it can also be assumed that all final concepts fulfill the environmental, societal, and economic requirements. As previously mentioned in Section 6.4, the material used for the final concepts remains the same as the one used in the current product: Corten steel. Using this material ensures that the exterior designs will be able to withstand the expected wear and tear while also providing great weather resistance, ensuring a robust construction. Additionally, corten steel is a recyclable material (İsmailoğlu & Akkan Çavdar, 2024), which further increases the concepts fulfillment of environmental requirements. Although this is based on the assumption that there are no hazardous materials used to treat or paint the exterior, which could decrease recyclability. However, the Sustainable inspired concept had the option to either use real wood or a metal-based wood finish, which affects recyclability depending on the material actually used. If the concept were to use real wood, it would most likely require a surface treatment or be pressure-impregnated to avoid becoming worn out quickly. Since this causes the wood to contain chemicals, it can make the recycling process more complex as the chemicals are released. However, there are multiple existing recycling companies able to handle this kind of recycling in a safe manner. Therefore, using real wood has a negligible impact on recyclability. Using the metal-based wood finish would not have a significant impact on recyclability either, as there are already well established companies with this purpose, such as Stena Recycling (“Om oss — Stena Recycling SE”, n.d.).

Due to exterior design being a rather abstract and subjective subject, it is impossible to objectively justify the concepts. Therefore, the concepts were evaluated in a highly subjective manner and deemed to fulfill the societal requirements defined. Each design was deemed to be aesthetically pleasing and to harmonize with their surrounding environment. Removing some of the standard elements found on 20-foot shipping containers, such as the corrugated panels and large door handles, allows other design elements to come forward. The integrated LED-lighting on the Scandinavian and Sustainable inspired concept creates a warm and inviting ambiance, which could further increase the general acceptance of

BESS being placed in urban or public environments. Additionally, each design fits into multiple different environments, which increases the versatility of the BESS and extends the range of possibilities.

Based on the financial analysis and commercial assessment conducted, all final design concepts were deemed commercially and economically viable. The exterior design is only a segment of the entire BESS as a whole system from a financial point of view. Therefore, as the final concepts do not include any major changes in either material or color, and keeps the entire frame structure, the overall impact of the additional cost was deemed negligible. Which, in turn, further supports the economical viability and justification of all the concepts. The full commercial assessment is provided in Section 8.

6 Prototypes

This section describes the different types of prototypes used throughout this project in detail. The purpose of prototypes in this project was to convey the intended designs more easily, rather than perform any quantifiable or qualitative tests.

6.1 Concept Gate

Three small prototypes, one of each design concept, were created for the concept gate to provide an interactive experience for the viewers during the Concept Gate presentation. The prototypes were created using three 3D-printed 20-foot containers, estimated to be at a 1:17 scale, using an already existing file of a 20-foot shipping container (“Shipping Containers 20 and 40 foot! by jrobertfox - Thingiverse”, n.d.), which was then adapted in PrusaSlicer. The concept designs were then printed in the right size onto three A4 papers, cut out, and taped onto the outside of the 3D-printed shipping containers. The result is illustrated in Figure 23 below.



(a) Short side, door



(b) Top side



(c) Long side, mirrored on both sides



(d) Short side, ventilation

Figure 23: Prototypes for Concept Gate

These small prototypes were passed around to the audience during the Concept Gate presentation to provide a better understanding of the concepts. This choice was made because the pictures can be hard to see in full detail from further away from the projector screen, also depending on the room lighting, the pictures can become quite faded. Additionally, by adapting the design to a physical version of the shipping container, the audience was able to better understand the exact placement of all details.

6.2 Digital Prototypes

Digital prototypes illustrated in Section 6.4, were created using the CAD program Autodesk Fusion 360, with the purpose of visualizing the design concepts in greater detail and allowing realistic renders to be made. To create the renders, each part of the CAD model was assigned real materials such as metal, wood, and fluorescent light. The CAD models for each concept were created using the same base for the frame, the placement of fire hatches, and the long side door.

The renders created using Autodesk Fusion 360, were then used as input to Google AI Gemini to help create different user environments and product families for each concept. Initially, High-Dynamic-Range Imaging (HDRI) environment were supposed to be used for the renders. However, due to the lack of time and limited number of suitable HDRI options found, the team instead chose to use AI programs. For the Industrial inspired design, the environment represents a charging station next to a highway, where the design feels well integrated to its surrounding. For the Scandinavian inspired design, the environment represents a charging station located more central in a city. Lastly, for the Sustainable inspired design, the environment represents a parking lot with charging stations for passenger cars in a suburban area.

6.3 Final Exhibition

Three updated prototypes, one of each design concept, were made for the purpose of visualizing the design as a physical product at the exhibition in the Study hall. The prototypes were created in a 1:10 scale of the BESS, with as much detail as possible given the available resources.

The 1:10 scale was measured to roughly 250x240x600 mm, with regard to some bending in the metal due to heat. The prototypes were created by welding 10x10 mm square steel bars for the frames; see Figure 24a, and 1,5 mm sheet metal for all main sides and details; see Figure 24b. To ensure the frame would be sturdy and not break easily, it was fully welded together using a TIG welder. All details, such as doors and fire hatches, were first spot-welded to stay in place, and then all doors were fully welded to create a more seamless look after painting. The fire hatches and doors on the long sides were placed exactly the same on the three prototypes to create a similar baseline for all designs. The short side, dedicated to ventilation, was not welded on any of the prototypes to enable a power bank or electrical outlet to be put inside. This was to mimic the actual purpose of the BESS and to pull out charging cables for viewers to charge their phones during the exhibition. Since the available resources were limited, both in regard to material and time, the level of detail excluded adding door handles, as it would be very time consuming. However, since the purpose of the prototypes was to simply provide a realization of the renders, the team deemed this level of detail sufficient.



(a) Welded frames



(b) Frames with some of the sides

Figure 24: Process of welding

After all three prototypes were welded together, they were painted using spray paint. The process of painting is illustrated in Figure 25 below. The Industrial and Scandinavian inspired designs used a half-glossy light gray color, while the Sustainable inspired design used a half-glossy cream white. For the color blocking on the Industrial design, a dark and slightly metallic blue was used, see Figure 25c.



(a) First layers of paint



(b) Last round of paint



(c) Color blocking on Industrial done

Figure 25: Process of painting

After all prototypes were painted, the final details were added, such as drilled holes for the charging cables on all prototypes, and the wooden details were added to the Sustainable inspired design. The wooden details were made using a 5 mm plywood that was engraved with a stripe pattern using a CO₂ laser cutter and then cut into four different pieces. The pieces were then taped together on the backside and placed on top of the Sustainable inspired design. For the exhibition, a number of different props were used to create a fun and interactive exhibition booth. Pictures from the booth are illustrated in Figure 26 below. For the Sustainable inspired concept, plastic grass and a couple of toy cars were used. Additionally, a smaller piece of wooden panel with an LED strip was used to illustrate the actual look of the wooden panel to be used. For the Scandinavian inspired concept, a printout of rocks was used, and for the Industrial inspired concept, a printout of asphalt and a toy Volvo Truck was used.

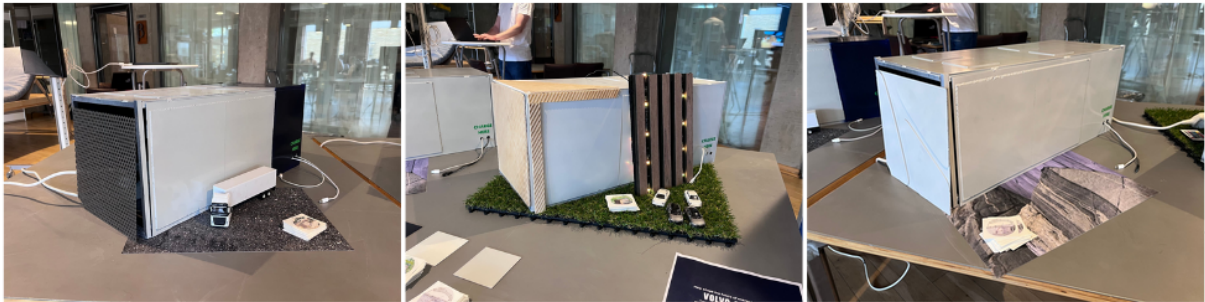
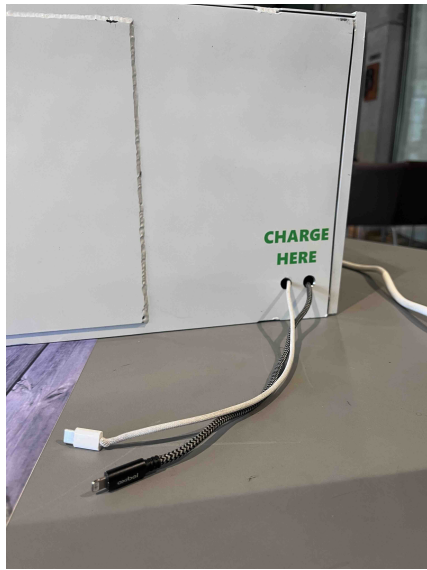
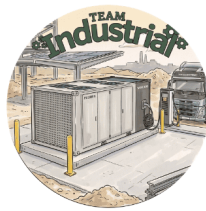


Figure 26: Final exhibition booth

To integrate an interactive part into the exhibition, electrical cords were placed inside each prototype to allow charging cables to be pulled through the pre-cut holes. This allowed visitors to actually charge their phones "using" the BESS, providing a simple and visual explanation of what a BESS is. The end result of the charging ports are illustrated in Figure 27a below. As an additional interactive element, stickers were created using the renders and adapted to look hand sketched using AI for each design concept. These stickers were made with the intention that visitors could get a sticker of the design they liked the most. The stickers are illustrated in detail in Figure 27b below.



(a) Charging port on prototypes



(b) Stickers

Figure 27: Charging port and stickers

7 Commercial Assessment

This section provides a detailed description and analysis of the commercial assessment conducted, which included evaluating the economic viability, conducting a sensitivity analysis, and identifying trade-offs.

7.1 Market Prerequisites

The commercial viability of the re-designed BESS concepts is supported by several favorable market conditions. The global market for BESS is experiencing strong growth, driven by increased electrification, expansion of renewable energy and a growing demand for grid flexibility and charging infrastructure (Wanner et al., 2026). Sustainability targets and regulatory initiatives across major economies further indicate sustained long term demand for energy storage solutions (IEA source). At the same time, the market remains highly regulated, requiring these products to comply with standards related to safety, fire protection, ingress protection, transportation and labeling, which all BESS products on the market must adhere to (Arvidsson et al., 2025).

Volvo Energy primarily operated within the industrial- and charging infrastructure segments, where customers include industrial facilities, fuel stations and charging network providers (Arvidsson et al., 2025). These customers invest in BESS units for reasons such as reducing energy cost, improving energy resilience and supporting electrification initiatives (Arvidsson et al., 2025). The market is currently dominated by large international competitors such as Alfen, Wärtsilä and Eaton, whose offering demonstrate a high degree of technical performance but rather generic exterior design and visual product identity. This creates an opportunity for Volvo Energy to differentiate in the market through exterior design, while maintaining excellence in product performance.

Moreover, products in this market are generally sold through business-to-business channels and are can be customized according to customer desires and requirements (R. Blom, personal communication, May 3rd, 2026). Given the continued expansion of the energy storage market and increasing deployment of commercial charging infrastructure, the market conditions indicate favorable prerequisites for commercializing an improved BESS offering that combines technical performance with enhanced usability, sustainability, and visual differentiation.

A central desired outcome of the project was that the re-design not only improves the visual aspects of the BESS units but also supports long-term economical success. According to Ulrich et al. (2020, p. 374), profitability is a key indicator of economic success, indicating that revenues exceed costs.

To evaluate the commercial viability and profitability of the re-designed concepts, a four-step commercial assessment was conducted. The analysis followed a structured approach and addressed both quantitative and qualitative elements. It consisted of a base-case financial model, a sensitivity analysis, a trade-off assessment, and an evaluation of qualitative factors, enabling both quantitative and qualitative aspects of the business case to be addressed and providing a comprehensive representation of its complexity. (Ulrich & Eppinger, 2011, p. 376). The quantitative analysis effectively evaluated measurable aspects of the business case, such as net present value (NPV), sales volume, and production costs. In contrast, the qualitative analysis served as a necessary complement, enabling an assessment of non-measurable, subjective characteristics of the BESS market, including interactions between the project and the firm, the market, and the macroeconomic environment (Ulrich & Eppinger, 2011, p. 375).

Moreover, Ulrich et al. (2020, p. 375) state several benefits of performing an economic analysis, such as providing input for business model selection, guiding go/no-go decisions, and supporting key decisions regarding operational design and development. In this project, the main value of the analysis is related to decisions concerning operational design and development. In theory, it could also support both business model decisions and go/no-go decisions. However, since Volvo Energy has already defined the business model for the BESS units, and since go/no-go decisions involve a broader set of considerations than the commercial value of the exterior design alone, the primary contribution of the commercial assessment is to provide supporting data and structured arguments for continued product development, as well as prioritization of activities within the development process. Furthermore, all data used in the commercial assessment is based on informed assumptions rather than actual figures from Volvo Energy. Since complete data was not available for all parameters in the financial model, a consistent assumption-based approach was applied across all inputs to ensure comparability and uniform accuracy across the assessment.

Although the project resulted in three distinct concepts, the financial analysis and commercial assessment indicated that the additional design elements contributed very similarly to the total cost across all three

concepts. While the financial model was developed separately for each concept, the results are presented here as a combined analysis, as this is sufficient to demonstrate the commercial viability of all three cases.

7.2 Cash Flow Analysis

The cash flow analysis evaluated the economic viability of the Volvo Energy BESS units under expected market conditions over a five year period, divided into quarterly intervals, using a standard 7 % discount rate, with NPV serving as the primary metric of interest. The analysis was conducted using a template in Google Sheets (A, Brahma, personal communication, April 15th, 2026), and all values were based on informed assumptions. In the model, sales volume and sales price were the main drivers of total revenue, while costs related to product development, equipment and tooling, production ramp-up, as well as marketing and support constituted the main components of total costs. Furthermore, the production cost per unit was divided into costs for the interior components of the BESS and the exterior design elements.

As the BESS units were planned to launch to the market in the coming years, the financial model mirrored this plan in terms of how each year was represented. Consequently, the first four quarters, corresponding to year 1, showed only cash outflows with no revenue generation. During this first year, product development costs accounted for the majority of expenses, totaling approximately 215 MSEK distributed evenly across the quarters. In addition, equipment and tooling as well as production ramp-up were estimated at 2 MSEK and 8 MSEK respectively. By the end of year 1, before any units had been sold, the project was estimated to have accumulated a cumulative cash flow deficit of approximately -212 MSEK.

During years 2-4, commercial operations were assumed to begin, which was mirrored in the model through an initial and steady quarterly increase in sales volume until reaching 50 units per quarter by the end of year 2. The assumption of a sales volume of 50 units per quarter, corresponding to 200 units annually, was made in accordance with Ulrich et al. (2020, p. 187), who stated that the approximate annual sales volume for a commercial airplane is below 200 units, while the corresponding volume for electronics manufacturing equipment typically ranges between 150-1500 units. As the BESS unit was identified as a product positioned between these categories, the annual sales volume was estimated at 200 units per year. At this stage, the units were estimated to be sold for 3 MSEK each, generating a quarterly revenue of approximately 151 MSEK. The production cost per unit was estimated at 2.12 MSEK, of which 0.32 MSEK represented the estimated cost of the exterior design, while marketing costs were estimated to remain at approximately 0.5 MSEK per quarter throughout the phase. Here, it is important to note that the cost contribution of the exterior design remained relatively small compared to the cost of the internal components, even when accounting for the additional design elements introduced in the redesigned concepts. The gross margin was estimated to 30 % , which was considered a reasonable assumption in relation to reported margins within the energy storage and battery systems industry (Aldona Krysiak-Adamczyk, 2025). During this phase, an important milestone was reached in year 3, quarter 4, as this was the point at which the business case was predicted to achieve financial break-even.

Year 5, which represented the final phase of the assessed time period, was characterized by increased market competition, introducing new market dynamics to navigate. This affected the business case through factors such as slight reductions in sales price and sales volume. Including these conditions in the assessment created a more realistic representation of likely future market conditions and provided valuable insights that could help guide future development and support proactive planning and decision-making of such developments.

To conclude, the overall financial base-case assessment resulted in a positive NPV of 224 MSEK, indicating a positive time value of money and showing that the long term cash inflows generated by the BESS units outweighed the present value of the cumulative investment and operational costs. From a long term profitability perspective, continued pursuit of the project was therefore estimated to add 224 MSEK in direct capital value to Volvo Energy, discounted at a 7 % cost of capital and based on the assumptions applied in the financial model.

7.3 Sensitivity Analysis

Based on the base-case financial model, a sensitivity analysis was used to evaluate how key commercial factors impact profitability. Building on the initial financial model, each parameter was evaluated using worst-case, baseline and best-case scenarios, in order to assess the robustness of the business case and

study how these changes affect the NPV (Ulrich & Eppinger, 2011, p. 381). This enabled an identification of the project's primary commercial risks and opportunities. As project value is influenced by both internal- and external factors (Ulrich & Eppinger, 2011, p. 381), both were explored, resulting in an evaluation of which parameters the business case is most sensitive to and providing further data to guide future development. In doing this, it was possible to test the resilience of the project by exploring extremes of critical factors.

The results displayed that the business case is highly sensitive to external factors such as sales volume and gross margin. A reduction in sales volume from 50 to 25 units per quarter decreased the NPV from 224 MSEK to approximately 13 MSEK, corresponding to a decline of over 94 percent. Similarly, reducing the gross margin from 30 percent to 15 percent resulted in a negative NPV of approximately -53 MSEK, indicating that margin erosion represents a critical commercial risk. In contrast, variations in manufacturing cost had a significantly smaller impact on profitability and NPV. Even under the worst case manufacturing scenario, the project maintained a positive NPV of approximately 198 MSEK, largely because, as the financial model shows, the redesigned exterior only adds a marginal increase in production cost compared to the current BESS.

Here, an interesting dilemma became apparent, namely that internal factors, which the development team have a high degree of influence over, has significantly less impact on profitability and NPV than external factors, which cannot be easily changed by the development team (Ulrich & Eppinger, 2011, p. 381). This suggests that reducing perceived quality in order to minimize internal costs such as manufacturing expenses is likely a suboptimal strategy. Instead, investing slightly more in design and quality to strengthen customer value perception would be a more favorable strategy, as this has the potential to improve sales potential and thereby support long term profitability. Furthermore, these results indicate that long-term commercial success is primarily dependent on achieving sufficient market adaptation and maintaining value-based pricing for healthy margins.

7.4 Trade-offs

Based on the sensitivity analysis, three trade-off rules were formulated to provide a greater understanding of the relative impact that different financial interactions had on NPV and to guide future decision-making (Ulrich & Eppinger, 2011, p. 384). In line with Ulrich et al. (2020, p. 387), the rules were formulated as changes in NPV per unit change of each factor. The resulting trade-off rules were as follows:

1. Approximately 2.6 MSEK change in NPV per 10% change in manufacturing cost.
2. Approximately 44.7 MSEK change in NPV per 10% change in sales volume.
3. Approximately 25.9 MSEK change in NPV per 1% change in gross margin.

7.5 Qualitative factors

While the base-case financial model and sensitivity analysis were effective in analyzing measurable aspects of the business case (Ulrich & Eppinger, 2011, p. 387), they were not able to sufficiently address the many subjective aspects that characterized the project. Since exterior design and visual expression were the primary focus of the project, and the perception of these is highly interpretive and perception-based, it was highly relevant to include an evaluation of qualitative factors in the commercial assessment in order to effectively analyze all relevant aspects of the redesign.

Ulrich et al. (2020, p. 389) define qualitative factors as those that are difficult to quantify due to their complex- or uncertain nature. To assess this, qualitative factors were analyzed from the perspective of interactions between the project and the firm,- the market,- and the macro environment (Ulrich & Eppinger, 2011, p. 389).

Interactions between the project and firm as a whole was analyzed under the assumption that firm profit would be maximized if project profit was maximized (Ulrich & Eppinger, 2011, p. 390). Here, relevant externalities were evaluated along with factors impacting the strategic fit between the project and the firm as a whole. When evaluating the interaction between the project and the market, competitors, customers and suppliers were addressed, as the actions and reactions of these are likely to affect project-specific factors like sales volume (Ulrich & Eppinger, 2011, p. 391). Lastly, the interactions between the project and the macro environment were addressed in terms of major economic shifts, government regulations and social trends (Ulrich & Eppinger, 2011, p. 391). The whole analysis was conducted by verbally addressing

and discussing each factor and interaction, and the analysis documented by text. Although all aspects of the qualitative factors were explored, a few stood out as most prominent. After evaluating each interaction and qualitative aspect, a number of key insights were identified, the findings were categorized into three major benefits associated with the subjective qualities of the redesigned concepts: Strengthening of the value proposition, market adaption and scaling, and organizational benefits.

In terms of strengthening the value proposition, the redesigned concepts' elevated exterior design increased perceived customer value, brand differentiation, trend resistance, and alignment with the MAYA principle.. Since the majority of competitor products currently looked very similar, a clear opportunity for differentiation through exterior design was identified. Therefore, each re-designed concept represents a unique opportunity for Volvo Energy to achieve clear brand differentiation and create favorable conditions for customers, stakeholders and the public to recognize their products based on their visual identity. Besides being beneficial for the company, customers and stakeholders would get advantages from this, as the improved aesthetics of the product would have greater potential to integrate more naturally into the environments in which customers of the BESS units would place them.

Moreover, since the concepts' distinct visual expressions were balanced with a minimalist design approach, they showed strong potential to align with the Most Advanced Yet Acceptable principle (MAYA principle), as the designs were innovative and modern enough to lead the market while still familiar enough to support immediate customer acceptance. Since many competitor products currently shared a similar visual appearance, the redesigned concepts also showed potential to become a benchmark within the industry, positioning Volvo Energy's BESS units as a recognizable and differentiated alternative within renewable energy and electrical vehicle (EV) charging infrastructure. Building on this, the minimalist visual expression placed the redesigned concepts in a balance between being visually distinct and sufficiently subtle, ensuring long term relevance and reducing the risk of the design becoming outdated as trends evolve. In this way, the concepts demonstrated trend resistance that could further strengthen customer appeal, as customers would be able to utilize the BESS units and benefit from their visual qualities throughout the product life cycle. Consequently, these qualitative advantages strengthened the business case by supporting the possibility of maintaining healthy gross margins, while the strengthened value proposition and increased brand differentiation also showed strong potential to contribute to higher sales volumes. In this way, the redesign of the product exterior address the most sensitive factors of the business case and contributes to making it more robust.

In terms of market adaptation and scaling, the improved visual expression showcased in the redesign enabled the BESS units to integrate more seamlessly into urban environments, creating the opportunity for the BESS units to be a modern component of the area where it is placed, allowing for placement in urban areas or sensitive environments, where traditional container-based solutions may have been rejected due to their limited aesthetic appeal. For example, gas stations that wished to keep their environmental initiatives relatively low profile in order to also accommodate customers using gasoline (Arvidsson et al., 2025) could implement BESS as green infrastructure without alienating parts of their existing customer base. Here, the open surface areas on the long sides of the redesigned BESS units address specific customer needs through their subtle visual expression and the opportunity for customers to apply their own branding or advertising to the exterior. Consequently, the re-design and improved visual expression of the BESS units shows promising to create broader market adaptation and scalability, potentially opening additional sales channels and sources of revenue while mitigating the risk of declining sales volume, which was identified as one of the most sensitive factors in the business case.

In terms of organizational aspects, the re-designed concepts create further opportunities long-term profitability beyond initial unit sales. Since the three concepts were developed with distinct and carefully considered design features, these features can be translated to similar products or across items in a product family. In this way, the re-designed concepts provide a foundation for Volvo Energy to develop and establish a unified design platform, by translating key visual elements across different sizes of BESS units, both existing and future product variants. By adapting these exterior design elements, greater scalability and flexibility could be achieved, ensuring that regardless of capacity, the BESS product portfolio remains visually coherent, recognizable, and adaptable to different customer needs.

Overall, the commercial assessment indicates that the redesigned BESS concepts show strong long term commercial potential. The financial base-case resulted in a positive NPV of 224 MSEK, despite significant initial investments and delayed revenue generation. The sensitivity analysis showed that profitability is highly dependent on sales volume and gross margin, while manufacturing cost had a comparatively small impact due to the limited additional production cost introduced by the redesigned exterior. This suggests

that investing in design and quality to strengthen customer value perception is likely a more favorable strategy than minimizing manufacturing costs. Furthermore, the qualitative assessment identified several strategic benefits related to strengthening the value proposition, improving market adaptation, and enabling future scalability. Altogether, the results indicate that the redesigned concepts have strong potential to create both financial and strategic value for Volvo Energy.

8 Exploitation plan

This section describes the strategy for commercializing the newly developed exterior design concepts for Volvo Energy's BESS unit. It outlines the market strategy, the continuous development requirements, and the financial strategies required to successfully introduce these designs to the market.

8.1 Commercial Strategy and Target Market

The commercialization strategy revolves around launching the redesigned BESS exteriors as the next generation standard and a market-leading premium offering. The product is highly commercial, as the exterior design not only acts as protection but also serves as a strategic tool for brand differentiation, public acceptance, and environmental integration.

Target Customers

The product is aimed to Business to Business customers, primarily charging infrastructure operators, commercial facility owners, and industrial operations. Leading users include mobility service providers providing electric vehicle charging in public or semi-public environments.

Alternatives and Consequences

The main alternative is to maintain the current aesthetic with a standard white container. The consequence of this is a missed opportunity for market differentiation and an increased risk of social resistance in urban environments. Another alternative explored was to implement a completely new modular architecture, such as a truss frame or a system based on a base plate with a frame. Although this is innovative, it would require a huge rework of the current production logistics.

Preferred Alternative

The chosen strategy is to implement the new exterior design themes by using the existing modified 20-foot container as a base. This prioritizes aesthetic innovation while minimizing disruption to the current assembly line in Skövde.

8.2 Market and Sales Plan

To effectively reach the market, the sales plan leverages Volvo Energy's existing ecosystem and strategic focus areas.

Customer Segments

The primary segment is public charging infrastructure and heavy-duty vehicle charging, as this sector exhibits a high sensitivity to visual quality and public integration. Secondary segments include large-scale commercial and industrial applications.

Product Variants

the BESS units are proposed to be offered as a unified product family based on a selected and refined exterior design language. To demonstrate the scalability of the design, the concepts have been visualized in several different physical sizes, including the standard 20-foot container, a 10-foot version and a smaller modular unit. This proposed flexibility would create a portfolio that can be fully customized to the customer's specific energy needs and available space.

Sale Channels

Sales will be made via direct Business to Business channels, leveraging the Volvo Group's extensive global network and supply chain. Strategic joint ventures, such as the Milence partnership for European high-performance charging networks, will serve as key distribution channels.

Customer Offerings

The BESS units are designed to act as a value-adding premium solution that goes beyond basic energy storage. By deliberately offering clean and uninterrupted surfaces on the long sides of the unit, customers are provided with an integrated platform for their own branding and marketing. This transforms the product into an active branding tool and creates direct business value by facilitating deployment in urban and commercial environments.

8.3 Continuous Development Plan

Transferring these concepts to active production requires a structured follow-up plan to ensure that all technical and aesthetic requirements are thoroughly verified.

Required Actions

The most critical next step is aerodynamic and thermal testing. As the new concepts consolidate ventilation onto a single side, physical testing is required to verify that internal airflows and cooling performance remain optimal. In addition, the exact material procurement, such as finalizing the selection of the Meta-Wood supplier for the sustainable-inspired concept, must be finalized.

Resources and Organization

The further development will be handled by Volvo Energy's internal engineering team. The final integration and assembly will be centralized to the production facility in Skövde.

Styling and Follow-up

A dedicated industrial design team must oversee the physical prototype development to ensure that the aesthetic integrity of the digital CAD models is strictly maintained during the transition to actual production.

8.4 Financial Plan

The financial strategy is based on the cash flow analysis, which indicates a very profitable business case for the new exterior design.

Financing Sources

As a subsidiary of the Volvo group, Volvo Energy benefits from strong financial backing. The initial product development, estimated at SEK 215 million, together with costs for equipment, tooling, and production ramp-up, will be financed internally through Volvo Energy's dedicated R&D budget.

Financial Outlook

The exterior design adds only a marginal cost, estimated at SEK 0.32 million per unit, to the total production cost. With an estimated sales volume of 200 units per year and a gross margin of 30 percent, the project is expected to break even financially by the end of the third year. The positive net present value of SEK 224 million strongly justifies the internal capital investment required to bring these exterior concepts to market.

9 Discussion

This section presents a critical discussion of the findings and results of this project. This section evaluates the credibility and validity of the outcomes, examines and analyzes the decisions made, and reflects on the choices made throughout the project.

9.1 Process

The progression from initial concept generation to detailed concept development highlighted the balance required between aesthetic identity and technical viability. By mapping distinct thematic directions and systematically filtering out initial configurations based on manufacturing complexity, transport constraints, and alignment with the Volvo design language, the team maintained necessary momentum while preserving the core elements that formed the final three concepts.

The subsequent technical analysis of alternative solutions and critical systems brought forward key structural and environmental insights. Exploring modular architectures, such as the truss-frame and base-and-pillar systems, revealed valuable long-term strategies for optimizing manufacturing, assembly and maximizing customization flexibility. Furthermore, addressing the strict technical demands of a large-scale BESS enclosure forced a critical integration of design and function. Issues regarding single-sided ventilation aerodynamics, solar heat absorption limits on palette choices, and material innovations, such as the potential implementation of MetaWood to resolve fire risks and thermal expansion mismatch, demonstrated that the exterior styling remains dependent on engineering principles. Because a typical BESS unit maintains a fixed operational lifespan of approximately 15 years, exterior design choices remain highly viable, serving as a documented foundation for future product architecture innovations at Volvo Energy.

9.2 Final Concepts

The final design concepts presented in Section 5.4 successfully aligned with the project scope while remaining within the constraint of using a standard 20-foot shipping container frame. By keeping standard structural elements, such as corner castings, forklift pockets, and long side doors the concepts remained compliant with regulatory standards. Additionally, by avoiding internal changes related to the overall performance and keeping the fire hatches, the concepts remained compliant with safety regulations. Replacing the classical corrugated steel surface with flat panels completely shifts the entire character of the product, creating a modern and clean look. Moreover, replacing the white finish creates significant differentiation within the competitor field. The light gray and cream white base colors preserve the ability for the exterior to reflect heat, while the dark blue may not be as effective. To accurately assess this property, testing and simulations should be conducted to evaluate exactly how much the color change affects overall performance. However, the centralized ventilation focused to one short side also brings potential risks and trade-offs. This limitation could heavily impact the air flow and cooling efficiency and require internal changes to be made. This was, however, outside of the scope of this project and was not further explored. Therefore, assuming full compatibility cannot be fully supported without further simulations and testing. The Industrial inspired concept further elevates the visual look with the color blocking to visually separate the internal sections. The Scandinavian inspired and Sustainable inspired concepts efficiently soften the harsh industrial feel of the product with their soft, nature-inspired design elements. However, the wooden panel used on the Sustainable inspired concept was identified as a potential fire hazard. It was noted that using a flammable material for the exterior could potentially violate fire and safety regulations. Due to the rigid safety regulations, the probability of an external fire spreading to the interior exhibits a low probability. Nevertheless, this could cause significant harm to the reputation of Volvo Energy. Therefore, using the metal-based wood finish eliminates the risk of an exterior fire. Furthermore, it is highly probable that using a metal-based wood finish would require less maintenance, as it is more durable and weather resistant.

Based on the subjective justification presented in Section 5.9, the final concepts successfully fulfill all requirements. By using the standard low-alloy corten steel, recyclability of a significant amount of the material used in the product was ensured. However, this assumed all parts were able to be separated and properly sorted, and as previously mentioned, no hazardous materials or contents. This was not further explored, therefore, the assumption requires a complete Product Life Cycle Analysis (LCA) to identify potential contradictions. Furthermore, evaluating the final concepts in a subjective manner may

have skewed the assumptions in favor of the concepts. However, due to exterior design not being as quantifiable as a tangible product, this approach emerged as the most practical strategy for this project.

9.3 Market Analysis

The BESS market is characterized by high technical and regulatory maturity but low product differentiation in exterior design, creating an opportunity for innovations within design as the market evolves.

Currently the market is dominated by well-established, large companies with access to strong resources. Competitor analysis shows that the majority of companies offer solutions that are highly similar in exterior design, based on standardized containers and painted white. This results in competition being driven mainly by cost and performance rather than user experience or product identity. This indicates a market where aesthetic design is not yet a strong competitive factor, creating both opportunities and challenges for Volvo Energy's development of BESS. Since Volvo Energy is part of a much larger scale company, Volvo Group, that is well-established and has resources, this gives them an easier way to scale up to better compete with the existing products. However, it also creates challenges because Volvo Energy is new to the BESS market compared to many of the other companies, which requires them to make greater efforts to establish themselves on the market.

The market analysis shows a high degree of standardization for BESS products. The requirements and regulations regarding safety, function, and transport create constraints which lead to a converging functional and aesthetic design. This gives Volvo Energy opportunities to create a product that differs from competitors and establish a new standard for BESS when it comes to visual design, brand identity, and integration into modern environments. However, the opportunities are challenged by the strict requirements which are shown in parts of the final products such as the corner castings and forklift pockets remaining standardized for transport. The new exterior designs with added features also have to be justified as profitable to motivate implementing aesthetic changes instead of keeping the old design. Another possible reason for the high similarity between the BESS products is that the market is still in a relatively early stage. Because the market has not yet reached a stage where designs start to differentiate and aesthetic aspects become more important competitive factors. Therefore, timing should be considered and analyzed for the implementation of the new designs in relations to market analysis. When launching a new and differentiated product in a market with high conformity, timing is crucial, as the market must be ready to accept the change.

Another factor affecting the market readiness of the product is the increasing societal integration of BESS. Although the BESS market is technologically mature, aesthetic design and user interaction aspects are less developed. Combined with the increasing integration of BESS into local communities and public environments, this creates a gap in the market with opportunities for differentiation through visual identity and usability that better aligns with public perception and acceptance.

9.4 Patent Analysis

From a patent perspective, the choice to use proven mechanical standard solutions was advantageous, as it allowed full focus on aesthetic expression instead of moving parts. Although applying for functional patents is not relevant, the concepts clearly distinguish themselves from ordinary standard containers by their unique way of hiding the necessary ventilation behind specific surface structures. This distinctive exterior justifies an application for design protection to prevent visual copying by competitors.

9.5 Manufacturing Methods and Adaptability

A well-developed exterior design not only functions as protection but also adds direct value to the product offering. By offering a BESS with an attractive exterior, the threshold for placing the units in public or urban environments is lowered, for example, at public charging stations. A conscious design significantly facilitates the customer's permit processes because it faces less aesthetic resistance from society and the general public. In addition, the exterior design serves as a strong tool for the customer's brand building by signaling green innovation and premium quality.

From a resource perspective, the concepts are well in line with Volvo Energy's established business model, which is already based on circularity. To further reduce resource consumption and strengthen the sustainability offering, the entire unit can be manufactured from recycled steel to the greatest extent possible, tying the visual identity of the product directly to its environmental purpose.

9.6 Requirements Specification

By offering more than just functional protection for the batteries, the new design concepts create real added value for the customer. This means that the previously rather anonymous exterior can now function as a strong and active tool for companies' branding and visual identity.

However, it is of utmost importance that these aesthetic additions are realized without compromising the strict technical requirements. A clear example of this is the aesthetic components that have been added to the unit's ventilation. As the basic ventilation itself has been preserved, a key goal has been that the new external additions must not, in any way, impair the existing protection. Instead, the design is carefully crafted to ensure that rain and snow continue to be channeled away effectively and safely, maintaining the system's necessary IP rating. In addition, strict adherence to UN standards has ensured that standardized lifting components and mandatory hazard markings remain fully visible and accessible during both transport and operation.

As the market for BESS units grows and these units are increasingly placed in or near public environments, the requirements for the systems are also changing. It is simply no longer enough for the industry to focus solely on optimizing internal battery technology. By skillfully balancing uncompromising physical security with adaptable aesthetics, new concepts help these devices integrate responsibly into their local environment. In this way, the design meets not only the technical needs, but also the rapidly growing ethical and social expectations for a modern green infrastructure.

9.7 Survey

The survey successfully fulfilled its exploratory purpose of obtaining an understanding of how the different concepts were perceived. The empirical result indicated an overall positive reception that closely aligned with the underlying design intent of each concept. However, as previously mentioned, the survey received ten responses in total. This was the result of the survey being distributed late in the project due to time limitations. This resulted in the gathered data lacking the statistical power to accurately represent the entire target population. While the survey was conducted to quantitatively gather data, to which Ulrich et al. (2020) recommends a larger sample size, it is also mentioned that the sample size may deviate depending on the desired data to be collected. Nevertheless, it is also noted that the sample size "should be large enough that the team's confidence in the results is high enough to guide decision making" (Ulrich et al., 2020, p. 174). In comparison to the entire target population, ten is statistically negligible in contrast. Therefore, the collected data may be insufficient to properly and accurately assess the public perception.

9.8 Prototypes

The prototypes created prior to the concept gate were effective in demonstrating the designs to the audience, and their small size made them easy to carry and pass around. They created an interactive experience for the listeners and allowed a closer look at the concept designs. However, the printed paper became quite damaged, and it was initially hard to make the tape stick to the plastic surface. Additionally, as illustrated in Figure 23 in Section 6.1, the 3D-printed container is not actually visible due to the paper. Therefore, it may not have been necessary to print scaled down 20-foot containers, as it did not add any further visual value. Instead, creating small container-like models using other materials, such as cardboard, may have been sufficient.

The digital prototypes created using CAD were heavily used throughout the later part of the project when the final concepts were decided. The CAD models were exceptionally effective in demonstrating the functions of different parts, such as LED-lighting and door openings. Using CAD to create digital prototypes is generally highly cost-effective by eliminating physical material consumption during development. Because the prototype was completely digital, changes could easily be applied at any given point without any additional costs. Furthermore, adding changes and manipulating a digital prototype is less time consuming than with a physical prototype. This allowed the team to fully explore multiple options without delaying the whole project. By assigning materials in the CAD program, the 3D-models provided high-fidelity visualizations that physical prototypes may not have been able to provide to the same extent. In addition to the renders created in the CAD program, the renders created using Google AI Gemini were also a significantly effective visual tool. Similar to using a CAD program, using an AI program was both cost effective and time efficient while still achieving high quality renders. The renders were a great way to demonstrate the different use cases for each concept and provided great support to

justify the design choices made. However, in the AI rendered figures illustrated in Section 5.4, there are a few noticeable mistakes in the environment or on the concepts. For example: the spelling of signs and the depth perception. Although these are insignificant mistakes that have a negligible impact on the overall render, they are details that AI did not notice. More critically, by using the designs as inputs for public AI models introduces several risks, as this compromises design exclusivity. Therefore, the design is no longer fully protected. Therefore, any competitor may be able to access them and copy the design, creating their own competitive advantage. To avoid this, it could be preferable to use HDRI environments to create renders within the CAD programs.

The concepts created for the final exhibition allowed an interactive experience for the visitors by demonstrating both the final design concepts and the function of the BESS. In comparison to the smaller 1:17 scale used for the first prototypes, the large 1:10 scale offered an enhanced visualization of each concept. However, because these prototypes were created using sheet metal and steel bars, each prototype became quite heavy and hard to move around. With an average of 14 kg per prototype. Substituting steel for aluminum or wood could have reduced the weight. However, using aluminum would have been harder to weld as it requires a different welding method than steel. It would also have made them less structurally stable since aluminum is easier to bend due to material properties. While using wood may not have been more time consuming, screws and glue would have been required to assemble the prototypes. This could have caused visual marks and details that may have affected the end result. Additionally, depending on the surface of the wood, streaks and marks may have been visible under the paint. Similar to aluminum, wood is not as structurally stable as steel and would have been easier to break. The decorations used for the final exhibition with the prototypes effectively created an interactive environment for the visitors.

9.9 Commercial Assessment

The commercial assessment indicates that the redesigned BESS concepts have strong long term potential, and several relevant insights can be derived from the results.

A central insight from the assessment is the imbalance between internal and external value drivers, where external factors such as sales volume and gross margin have a substantially greater impact on NPV than internal factors such as manufacturing cost. This creates a paradox, as development teams primarily control internal cost structures and product attributes, while overall financial performance is largely determined by market-driven variables. Although this may introduce challenges in product development, it also highlights the importance of prioritizing actions that enhance market-facing value rather than focusing solely on internal cost optimization. Furthermore, the qualitative assessment indicates that internal design decisions can indirectly influence these external drivers. In particular, the improved exterior design shows strong potential to enhance perceived value, support differentiation from competitors, and address customer preferences. These effects may, in turn, contribute to higher sales volumes and stronger pricing power. This supports the conclusion that exterior design can carry meaningful commercial value beyond purely aesthetic considerations.

Another key insight is the clear implication of a shift toward value-centric rather than cost-centric development. The results suggest that investing in the visual and experiential aspects of the product can be economically justified when such investments strengthen perceived value and reduce sensitivity to changes in sales volume and margin. However, this requires that the development team has access to relevant customer insights and a clear understanding of how design choices translate into perceived value. By grounding design decisions in customer needs and market expectations, product development can be more closely aligned with user-centric- and market-driven innovation rather than internal efficiency optimization alone. Moreover, this may introduce cross-functional tensions within larger organizations, where engineering, sales, and design functions may have different priorities and success metrics. Therefore, effective alignment between stakeholders is essential to ensure that exterior design is understood not only as aesthetics, but as a strategic lever for commercial success. Once the functional- and performance-related aspects of the product have been optimized, exterior design can serve as a differentiating factor that elevates the product from a standard offering to a market-leading solution that delivers yields perceived customer value.

The findings of the commercial assessment are based on a simplified financial model and a number of assumptions regarding sales development, pricing, and cost structures over time. In reality, market dynamics are likely to be more complex, and factors like potential competitor reactions and uncertainty regarding customer adoption, are not fully captured in the model. Additionally, the qualitative assess-

ment, which is based on interpretative analysis, introduces a degree of subjectivity in how benefits and impacts are evaluated. As a result, the assessment and the conclusions drawn from it should be interpreted as directional insights rather than precise predictions.

10 Recommendations

The exploitation plan in Section 8 establishes a strategic and commercially viable framework for the introduction of the new design to the current BESS market. Volvo Energy should capitalize on the opportunity for market differentiation by applying the new exterior design to the existing 20-foot container. Not only would this create a competitive advantage, but it would also increase the social acceptance of BESS in urban environments. As previously mentioned, this strategy would prioritize aesthetic innovation while minimizing disruption to the current assembly plant. By preserving all standard structural elements and only changing the exterior panels, no extensive changes to the assembly line would be required. Because Skövde predominantly handles final assembly, they do not manufacture components in-house. Therefore, the likelihood of requiring major investments for the assembly plant in Skövde is low. In turn, this paves the way for a significant advantage while also being cost-effective.

Furthermore, there are multiple approaches to leverage Volvo Energy's existing ecosystem and strategic focus areas. Maintaining the primary market segment as public charging infrastructure provides a significant opportunity for market growth, as this sector exhibits a high potential for public integration. By establishing their position here, Volvo Energy has the potential to become the generalized trademark through increased product visibility. Offering a unified product family demonstrates scalability and opens up opportunities for modularization. This increased flexibility widens the possible market segments by allowing customization and offering the same functionality on a smaller scale. Additionally, by offering clean and uninterrupted surfaces on the long sides of the unit, both Volvo Energy and customers are provided with an opportunity for branding and marketing. This creates direct added business value that may influence customers to choose Volvo Energy over other competitors.

Finally, to successfully transfer these concepts to active production, a systematic and strategic development plan is recommended. First, comprehensive aerodynamic and thermal testing is advised to analyze how the internal cooling and airflow are affected by the modified ventilation. This is critical in order to evaluate and validate the actual feasibility of the designs. Additionally, for the Sustainable inspired concept, a finalized decision on whether real wood panels or metal-based wood panels will be used is required. To properly conduct a material analysis, an LCA combined with simulations, testing, and a risk analysis is required. This allows for extensive evaluation of potential weaknesses and strengths to be identified, providing essential data.

11 Concluding Action Plan

To further continue the development of this project, the following concluding action plan should be implemented.

The technical validation of the concepts should be further explored and properly assessed by the team. As stated in Section 9, aerodynamic and thermal testing should be conducted to analyze how internal performance is affected by having ventilation focused on only one short side. Furthermore, simulations and environmental testing should be conducted to evaluate exactly how much the heat absorption changes between the original white, cream white, light gray, and dark blue panels. This analysis is necessary to identify any compromises to the internal thermal regulation.

The team should also further explore, alongside Volvo Energy, the modular architecture mentioned in Section 5.2, creating functional digital prototypes to evaluate feasibility. Furthermore, design modularity and how the designs could be translated into different scales while keeping their visual identity should be evaluated. Then, analyzed how this could provide leverage to Volvo Energy on the BESS market. Additionally, a financial analysis and commercial analysis should be conducted for the entire product family to evaluate economic feasibility and impact.

To further validate public acceptance, the required safety labeling should also be added to the designs to accurately depict how the designs would look in reality. The new visualizations should then be used to conduct another concept evaluation, as the overall perception of the designs may shift completely. Furthermore, the initial survey conducted should be expanded and distributed to a substantially larger sample size to reduce sampling errors. Additionally, a qualitative analysis of each concept should be conducted to provide a deeper and more nuanced understanding of *why* a concept is perceived in a certain way.

12 Conclusion

This project explored how the exterior design of Volvo Energy's BESS can be developed to balance aesthetic value with regulatory and functional requirements. Through customer, user and stakeholder analysis, market analysis, benchmarking, concept generation, development, and evaluation, opportunities and constraints for the exterior design were identified and translated into design solutions.

The results show an indication that the current BESS market is highly standardized, where products are differentiated through technical performance rather than exterior design. Simultaneously, BESS are increasingly being deployed in public environments which highlights a growing demand for visual integration and brand identity in combination with safety and usability. This creates the potential for visual, exterior design to become a significant competitive factor as the market matures. The findings suggest that exterior design may contribute to creating customer value beyond technical performance alone, creating an opportunity for Volvo Energy to develop a product that differs from competitors in an increasingly standardized market.

The project resulted in three different approaches to balancing aesthetic and functional requirements through the Sustainable, Industrial and Scandinavian concepts. The concepts provide different directions and applications for Volvo Energy as well as the customer and user groups, demonstrating how exterior design can contribute to both visual identity and improved user experience and integration into surrounding environments.

Overall, this project concludes that exterior design can contribute to values beyond aesthetics and that BESS can become more than just a technical product. It shows that exploring the design space and creating a new visual design can serve as a strategic tool for differentiation. By re-imagining the design aspect of energy storage systems and how they interact with their surroundings, the project contributes to Volvo Energy's ambition of creating modern and recognizable energy storage solutions for the future.

Appendix

A Competitor solutions



Figure 28: Fluence, GridStack Pro 5000, (**FluenceEnergySafety**)



Figure 29: Wärtsilä, Quantum 3, (**WartsilaTechnology**)



Figure 30: Sungrow, PowerTitan 2.0, (**SUNGROW**Sungrow**SUNGROW**)



Figure 31: BYD, MC Cube, (**WITSolarPower****BYDPOWER**)



Figure 32: Canadian Solar, SolBank 3.0, (**E-storageSolBankStorage**)



Figure 33: Battlink, High-voltage On-grid Liquid-cooled ESS, (**BattlinkHigh-voltageBattlink**)



Figure 34: Trina, Elementa 2, (TrinasolarTrinasolar)



Figure 35: Tesla, Megapack 2XL, (TeslaMegapackTesla)



Figure 36: CATL, EnerC, (EvlithiumCATLSystem)



Figure 37: Wärtsilä, Quantum HE, (WartsilaWartsilaDensity)



Figure 38: Etic AG, Legion C20, (AGBESSProductsAGBESS)



Figure 39: Saft, Intensium Shift, (SaftIntensiumSaft4U)



Figure 40: Cummins, C2000B5ZE, C1500B5ZE, C1000B5ZE, C600B5ZE, (ADEPowerCumminsSystem)



Figure 41: Huawei, LUNA2000, (SolaricsHuaweiGmbH)



Figure 42: Volvo, PU2000, (**VolvoEnergyPU2000Energy**)



Figure 43: Eaton, xStorage Container, (**EatonEnergyEaton**)



Figure 44: Fluence, Gridstack Pro 2000, (**FluenceEnergySafety**)

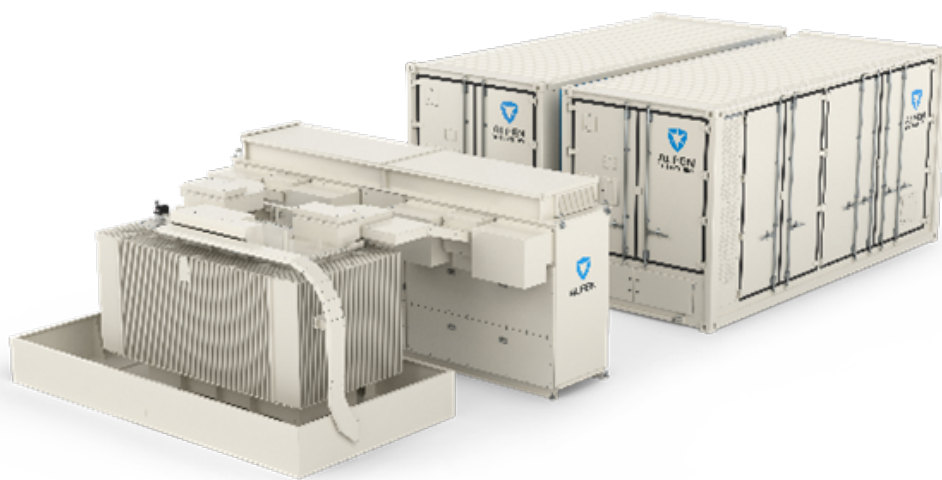


Figure 45: Alfen, TheBattery Elements 2.0, (**AlfenEnergyAlfen**)

B Datasheet - 40CrMoV13-9

General information

Designation ⓘ

Alloy special steel, 40CrMoV13-9, quenched and tempered, nitriding steel		
Condition	ⓘ	Quenched and tempered
EN name	ⓘ	40CrMoV13-9
EN number	ⓘ	1.8523

Tradenames ⓘ

Nitrodur 8523; Virgamet; Thyssenkrupp; Ovako; Fushun special steel; Basedo steel		
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Typical uses ⓘ

Die blocks, punches, aerospace components, molds for plastic materials, ships, vehicle components, guided missiles, weapons, railways, bridges, pressure vessels, machine tools, mechanical components with a large sectional size, mechanical gears, gear shafts, main axes, valve rods, connecting rods, bolts, nuts		
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Price

Price	ⓘ	* 15,5	-	24,4	SEK/kg
Price per unit volume	ⓘ	* 1,2e5	-	1,9e5	SEK/m^3

Physical properties

Density	ⓘ	7,7e3	-	7,8e3	kg/m^3
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Mechanical properties

Young's modulus	ⓘ	190	-	210	GPa
Specific stiffness	ⓘ	24,5	-	27,1	MN.m/kg
Yield strength (elastic limit)	ⓘ	750	-	810	MPa
Tensile strength	ⓘ	950	-	1,15e3	MPa
Specific strength	ⓘ	96,7	-	105	kN.m/kg
Elongation	ⓘ	11	-	15	% strain
Tangent modulus	ⓘ	2,95e3			MPa
Compressive modulus	ⓘ	* 190	-	210	GPa
Compressive strength	ⓘ	* 750	-	810	MPa
Flexural modulus	ⓘ	* 190	-	210	GPa
Flexural strength (modulus of rupture)	ⓘ	* 792	-	959	MPa
Shear modulus	ⓘ	* 73,7	-	81,5	GPa
Shear strength	ⓘ	610			MPa
Bulk modulus	ⓘ	* 151	-	167	GPa
Poisson's ratio	ⓘ	0,29			

Composition overview

Compositional summary ⓘ

Fe94-95 / Cr3-3.5 / Mo0.8-1.1 / Mn0.4-0.7 / C0.35-0.45 / V0.15-0.25 (impurities: Si<0.4, P<0.035, S<0.035)		
Material family	ⓘ	Metal (ferrous)
Base material	ⓘ	Fe (Iron)

Composition detail (metals, ceramics and glasses)

C (carbon)	ⓘ	0,35	-	0,45	%
Cr (chromium)	ⓘ	3	-	3,5	%
Fe (iron)	ⓘ	93,5	-	95,3	%
Mn (manganese)	ⓘ	0,4	-	0,7	%
Mo (molybdenum)	ⓘ	0,8	-	1,1	%
P (phosphorus)	ⓘ	0	-	0,035	%
S (sulfur)	ⓘ	0	-	0,035	%
Si (silicon)	ⓘ	0	-	0,4	%
V (vanadium)	ⓘ	0,15	-	0,25	%

Figure 46: Material properties of 40CrMoV13-9

Impact & fracture properties

Fracture toughness	i	* 21	- 156	MPa.m ^{0.5}
Toughness (G)	i	* 3,68	- 73	kJ/m ²

Thermal properties

Melting point	i	1,42e3	- 1,46e3	°C
Maximum service temperature	i	480		°C
Minimum service temperature	i	* -65	- -35	°C
Thermal conductivity	i	39	- 42,6	W/m.°C
Specific heat capacity	i	470		J/kg.°C
Thermal expansion coefficient	i	11,1	- 13	µstrain/°C
Thermal shock resistance	i	294	- 359	°C
Thermal distortion resistance	i	3,1	- 3,71	MW/m
Latent heat of fusion	i	260		kJ/kg

Electrical properties

Electrical resistivity	i	19	- 22,1	µohm.cm
Electrical conductivity	i	7,8	- 9,07	%ACS
Galvanic potential	i	* -0,44	- -0,36	V

Processing properties

Metal casting	i	Unsuitable		
Metal cold forming	i	Excellent		
Metal hot forming	i	Excellent		
Metal press forming	i	Excellent		
Metal deep drawing	i	Limited use		
Machining speed	i	15,2		m/min
Weldability	i	Poor		
Notes		Preheating and post weld heat treatments may be required		
Carbon equivalency	i	1,21	- 1,6	

Durability

Water (fresh)	i	Excellent		
Water (salt)	i	Excellent		
Weak acids	i	Acceptable		
Strong acids	i	Limited use		
Weak alkalis	i	Excellent		
Strong alkalis	i	Limited use		
Organic solvents	i	Excellent		
Oils and fuels	i	Excellent		
Oxidation at 500C	i	Acceptable		
UV radiation (sunlight)	i	Excellent		
Galling resistance (adhesive wear)	i	Acceptable		
Flammability	i	Non-flammable		

Corrosion resistance of metals

Stress corrosion cracking	i	Not susceptible		
Notes		Rated in chloride; May be susceptible in halide, ammonia, nitrogen, acidic, caustic, carbonate environments		

Figure 47: Material properties of 40CrMoV13-9

C Survey

Research Survey Form

<https://forms.gle/e3McNAaSDTYmKvG1A>

D Survey Result



Figure 48: Result from Survey - Industrial



Figure 49: Result from Survey - Scandinavian

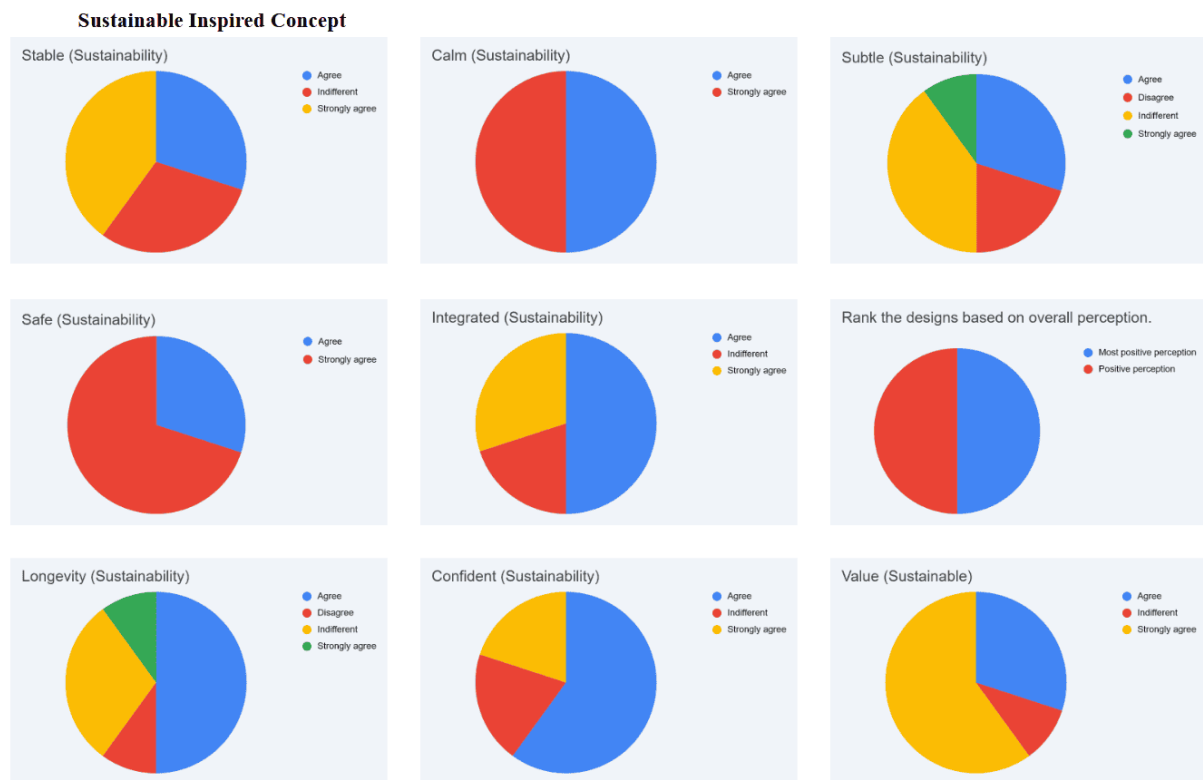


Figure 50: Result from Survey - Sustainable

E Reflection Report



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MPP126 - Reflection Report

Volvo Energy: Exterior Design of Battery Systems

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1 Introduction

This section gives an overview of the project and its context. It begins with a brief background, which highlights the emerging potential found within Battery Energy Storage Systems (BESS) and the urgency to transform the design to a more modern solution. The aims and research questions are then presented, defining the objectives of the project. Lastly, the scope is established to define the project boundaries.

1.1 Background

In today's rapidly evolving technological landscape, increasing electrification and growing sustainability demands are driving the need for smarter and more resilient energy solutions. In this landscape of technology, Battery Energy Storage Systems (BESS) have emerged, showing great potential for enhancing energy grid stability, enabling higher penetration of renewable energy and functioning as charging infrastructure (Arvidsson et al., 2025). These units have several applications for different sectors of society, which means that they will be placed in public places (Anastasovski et al., 2024). Therefore, there is a need for BESS units to exhibit well-developed exterior design, in addition to their essential technology and internal components.

According to Ulrich et al. (2020, p. 12) the product development process is a sequence of steps and activities that transforms a set of inputs into outputs, by conceiving, designing and commercializing a product. The phases in which these activities are performed are generally divided based on the nature of the project stage, such as ideation, concept development, and concept generation (Ulrich et al., 2020, p. 12). Moreover, this project has adapted the specific activities within each phase to fit the aim and nature of the project. The structured approach in the development process not only provided a clear, common direction for the end goal, but also enabled continuous development of knowledge throughout the process, as well as risk reduction in the process (Ulrich et al., 2020, p. 13).

1.2 Project Aim

The project has the following aim:

"Re-imagine and re-design the exterior design of the Volvo Energy BESS units into a discreet, clean and modern-looking energy storage solution that represents the Volvo brand and contributes positively to overall performance. "

1.3 Scope

The scope of the project is specifically focused on the exterior design of the Volvo Energy BESS units, positioning it's visual expression as a primary opportunity for differentiation among competitors. While overall technical performance remains essential for the product as a whole, this project emphasizes exploring the visual aspects and optimizing the exterior. Still, functional and technical requirements need to be fulfilled to justify any exterior- or visual feature, and serve to define the boundaries of the design space, ensuring alignment. Consequently, the prioritization of visual aspects does not treat functionality and design as separate elements, but rather emphasizes their integration. The exterior design is approached as a way to present the product's capabilities in a visually appealing, practical, and cost-effective manner, while also allowing for future adaptation and refinement. This approach was reflected in the choice and application of methods during the entire development process.

2 Methods

This section provides a detailed description of all the methods used in the project. The section is divided into two major phases of the design process: ideation and evaluation.

2.1 Ideation

The ideation phase utilized a structured approach to explore a wide design space while taking measures to avoid common development challenges and dysfunctions. The approach began with the establishment of an initial direction, and then moved on to employ a combination of individual- and team brainstorming methods, followed by multiple morphological matrices.

2.1.1 Initial Direction

In the early stages of the project, the initial direction was established using two methods: a requirements list and mood boards.

The requirements list was developed based on previous customer interviews and research (Arvidsson et al., 2025). Based on the collected data, the identified customer needs were translated into requirements. Since the product is an industrial energy storage solution, a large part of the requirements also consisted of strict industry standards and legal requirements (e.g., ISO standards for shipping containers and specific IP-ratings). These constituted the fundamental constraints of the project.

In parallel with defining the technical specifications, mood boards were used to define and clarify the aesthetic intention of the product. The purpose of using mood boards was to lay the foundation for the product's visual equity early on, which Ulrich et al. (2020, pp. 222–223) highlights as a central practice in industrial design to create a consistent and recognizable expression. To capture brand identity, which is another key goal in industrial design (Ulrich et al., 2020, pp. 222–223), and the desired perception of the product, mood boards were first created individually by the team members and then merged into a common mood board. The visual language aimed to, through central semantic values, both evoke the desired emotions and reflect the company's overall design philosophy (Ulrich et al., 2020, p. 216). The words chosen to describe these values were: *Safe, Stable, Longevity, Integrated, Calmly Strong, Quietly Confident, High Quality, Future-ready, and Sustainable*.

2.1.2 Brainstorming Methods

There are several methods for generating design solutions, both creative and rational (Almefelt, 2026, p. 79). The creative methods used in this project were Brainwriting 6-3-5 and SCAMPER.

Brainwriting

Brainwriting 6-3-5 is a brainstorming method where ideas are generated individually in written form (Wikberg Nilsson et al., 2021, p. 139). The method is originally structured with six participants but in this specific context, five team members participated. To begin the process, a clear question was defined: "How can a functional and safe BESS system be visually integrated and be pleasing to its environment?". Based on this question, each participant wrote down and drew three ideas each within five minutes. After this, the paper was passed on to the next participant who, for five minutes, added to the initial idea. This was repeated several times, and after about 25 minutes, multiple ideas had been generated and developed. Lastly, all ideas were collected, organized, and discussed among the team. During this process, specific guidelines were emphasized in accordance with recommendations from Wikberg Nilsson et al. (2021, p. 139) and the lecture on Systematic Design by Almefelt (2026, p. 79). Some of the guidelines were: avoid criticism, set a goal for the minimum number of generated ideas, do not exceed an hour, feed the team relevant questions, organize, classify, and evaluate ideas after the session.

This method was selected as a suitable approach for generating design solutions, due to its ability to produce a large number of ideas within a limited time in a structured way. Producing many ideas aligns with the guideline of setting a goal for a minimum number of generated ideas. The method also ensures that all participants are actively contributing and creating a free flow of ideas, which reduces the risk of one participant dominating the process, which aligns with the guideline of avoiding criticism. The structured timing of the method, lasting approximately 30 minutes, makes the process efficient, which is in accordance with the guideline of the process not exceeding one hour.

SCAMPER

SCAMPER is a creative method that is used to create new ideas by rethinking, rearranging or modifying existing solutions (Ulrich et al., 2020, pp. 133–134). First, a few ideas from Brainwriting were selected for further development with SCAMPER. Then, techniques such as substitute, combine, adapt, modify/magnify/minimize, put to other uses, eliminate, and reverse/rearrange were used to explore how the ideas could be changed or developed further (Ulrich et al., 2020, pp. 133–134). These ideas were then visualized through sketches and descriptions and then evaluated. During the implementation of this method, the above mentioned guidelines stated in the lecture on Systematic Design by (Almefelt, 2026, p. 79), were used.

This method was selected because it provided structure for improving and transforming existing ideas. By encouraging participants to modify parts of the concepts, it aided in exploring new solutions and relationships among ideas (Ulrich et al., 2020, pp. 133–134). SCAMPER supports continued idea generation by guiding participants through different actions, which also increases the number of developed ideas. This structure within the method also aligns with the guideline of feeding the team relevant questions, which ensures that the idea generation process does not stagnate.

2.1.3 Morphological Matrix

The conceptual design phase is a critical phase that requires rapid iterations to explore diverse concepts. To support this, the team used the Morphological Matrix (MM), a systematic method for forming total solutions by combining sub-solutions (Almefelt, 2026; Martinsson Bonde et al., 2022). This approach is functionally similar to the Concept Combination Table, which provides a structured way to consider combinations of solution fragments (Ulrich & Eppinger, 2011, p. 134).

The process began with a functional decomposition, utilizing a function-means tree to divide the primary function of the Battery Energy Storage System (BESS) into sub-functions. Through iterations, the final MM was organized into three primary categories, which were further divided into sub-hierarchies:

1. Functional: Heat management, fire safety, water leak prevention, transport/installation, service, modularity, vandalism protection, and access.
2. Appearance: Material, ventilation integration, lighting and overall form.
3. Perception: Brand expression, environmental integration, safety communication, and function interpretation.

Due to the high number of categories, the team used a prioritization system. "Starred" categories were addressed in the initial concept generation, see Figure 1. These categories included the most important features required to create a baseline. Without these, the overall performance would decrease, or the design would be incomplete, such as heat management, material, and general shape. While "unstarred" categories were moved to a secondary matrix for detailed development, see Figure 2. These categories were not important to the overall performance, but rather add-on features that could potentially increase the overall design and value. A second copy of the matrix served as a "storage box" for new ideas that surfaced during the process; see Figure 3. To manage interactions between categories, a color-coded post-it system was used on Miro to link solutions across categories. During concept development, specific "direction" paths (e.g., 5 red and 5 orange categories) were mapped through the matrix to synthesize cohesive system solutions. The color coding represents the two groups that mapped different concepts, red being group one and orange being group two, and the numbers in each circle represents the concept number. Additionally, the matrix served as a feedback tool during a visit to Volvo Energy, allowing stakeholders to provide targeted comments on specific sub-solutions.

Following the guidelines of Ulrich et al. (2020, p. 119), this structured approach prevented common dysfunctions, such as considering only a few alternatives or failing to explore entire categories of solutions. To maximize output, the team followed both internal and external search (Ulrich et al., 2020, p. 134) by suspending judgment, welcoming ideas that may have seemed infeasible, and utilizing graphical and physical media (Ulrich & Eppinger, 2011, p. 128). Both individual- and team brainstorming were conducted to generate a high volume of ideas while also building team alignment and refining ideas together.

[illegible]

Figure 1: Main Morphological Matrix

[illegible]

Figure 3: Updated Morphological Matrix

2.2 Evaluation

The evaluation phase focused on narrowing down the concepts generated during the ideation phase, through systematic comparison and qualitative discussion. The concepts were first evaluated using a Pugh matrix, that focused on addressing both functional and aesthetic criteria. Then, the concepts were evaluated using The Value-method, which focused on assessing the visual aspects in a subjective manner.

2.2.1 Pugh Matrix

The Pugh matrix was used to narrow down the number of concepts generated during the ideation phase. According to Ulrich et al. (2020, p. 156), using the Pugh matrix is part of concept screening, which is used to quickly improve and reduce the number of concepts.

The Pugh matrix applied in this project included both functional and aesthetic criteria established by the team, as illustrated in Figure 4 below. The current exterior design by Volvo Energy was used as the reference to set a neutral baseline for the concept screening. Each concept was then evaluated against all criteria, where + meant higher performance, - meant lower performance, and 0 meant the same performance as the reference. However, due to lack of specific data at this point in the project, some criteria were marked Not Available (n/a). The use of this value was primarily motivated by the fact

that the scope of this project did not include the technical functionality of the BESS, which caused the previously mentioned lack of information. Additionally, this value was used to avoid influencing the result in a subjective manner. Due to the high number of n/a, it was concluded that this evaluation method was insufficient for evaluating the subjective criteria and other factors related to exterior design.

Thereafter, all concepts were given a net value based on the sum of all values where + was equal to +1, – was equal to –1, and 0 was equal to 0. The net value was then used to rank the concepts from 1 – 5, where concepts 1 and 4 were ranked at the same value, as illustrated at the bottom of Figure 4. As previously mentioned, the Pugh matrix is intended for concept screening, and in this particular case, it was not possible to reduce the number of concepts solely based on the result of this method. This was due to the high number of n/a caused by the lack of data and information and insufficient evaluation of subjective criteria.

Chalmers		Pughmatrix: BESS Exterior Design (Volvo Energy)				
Created: 2026-03-06						
Criteria	Concepts					
	PU2000	Concept 1 Sustainability	Concept 2 Scandinavian	Concept 3 Industrial	Concept 4 Sci-fi	
						
Technical						
Maintain 23 C as internal temp.	0	n/a	n/a	n/a	n/a	
Temperature regulation	0	n/a	n/a	n/a	n/a	
Ventilation - specify	0	n/a	n/a	n/a	n/a	
Protect from humidity and water.	0	n/a	n/a	n/a	n/a	
Protect from dust and pollution.	0	n/a	n/a	n/a	n/a	
Min. lifespan: 15 years	0	n/a	n/a	n/a	n/a	
Fulfill laws for fire safety.	0	n/a	n/a	n/a	n/a	
Fulfill CSC 8.	0	n/a	n/a	n/a	n/a	
Prevent bodily injury.	0	n/a	n/a	n/a	n/a	
User-centered/aesthetic						
Mediate positive associations.	0	+	+	+	0	
Mediate quality (worth the money).	0	0	+	+	0	
Mediate stability.	0	0	0	0	0	
Mediate robustness.	0	-	-	0	0	
Intuitive exterior design (interaction, UX)	0	+	0	+	+	
Modern design.	0	+	+	+	+	
High-tech design.	0	+	+	+	+	
Modular design.	0	n/a	n/a	n/a	n/a	
Exterior that enables temperature regulation.	0	0	0	0	0	
Scratch resistant exterior.	0	0	0	0	0	
Service modularity.	0	n/a	n/a	n/a	n/a	
Service availability.	0	+	+	+	+	
Installation availability.	0	n/a	n/a	n/a	n/a	
Exterior that facilitates cleaning.	0	0	+	+	0	
Different from competitor solutions exterior.	0	+	+	0	+	
Mobility opportunities after installation.	0	0	0	0	0	
Sustainability						
Environmental impact of exterior materials.	0	-	-	-	-	
Cost						
Manufacturing cost < 300 000 SEK	0	n/a	n/a	n/a	n/a	
Σ +	0	6	7	7	5	
Σ 0	27	6	5	6	8	
Σ -	0	2	2	1	1	
Net value	0	4	5	6	4	
Ranking	5	3	2	1	3	
Further development						
Decision	Further evaluation of all concepts.					

Figure 4: Pugh matrix

2.2.2 The Value Method

The Pugh matrix was effective for addressing some of the functional requirements for the concepts and for identifying the strongest two concepts among the four. However, it did not address the aesthetic requirements of more subjective nature in a satisfactory manner. To address this and enable further elimination, the Value method was used. According to Wikberg Nilsson et al. (2021, p. 234), the Value-method is a method used to evaluate concepts in a time-effective, systematic way. Addressing the advantages, limitations and unique elements of each concept allows for a comparable evaluation of the concepts. The method stimulates discussion in the development team, to provide a better understanding of each concept and how effectively they are in fulfilling requirements and the aim of the project. This method for evaluating can best be classified as a pros and cons approach to evaluation, as the strengths and weaknesses of each concepts was listed and discussed and choices were based on the opinions of the design team (Ulrich & Eppinger, 2011, p. 151). Since the exterior design was a strong focus for the project, addressing the visual aspects in a sufficient way was of highest priority. Therefore, the Value method

was a reasonable complement to the Pugh matrix, and enabled for sufficient evaluation of the aesthetic requirements, which are of a more subjective nature. The structured approach used when applying the method enabled equivalent evaluation of all included concepts. The process resulted in a clear output, where one of the concepts had significantly more limitations compared to the other concepts, which led to the elimination of that particular concept.

2.2.3 Commercial assessment

A central desired outcome of the project was that the re-design not only improves the visual aspects of the BESS units but also supports long-term economical success. According to Ulrich et al. (2020, p. 374), profitability is a key indicator of economical success, indicating that revenues exceed costs.

To evaluate the commercial viability and profitability of the re-designed concepts, a four-step commercial assessment was conducted. The analysis followed a structured approach and addressed both quantitative and qualitative elements. It consisted of a base-case financial model, performed in a templated Google Sheets document (A, Brahma, personal communication, April 15th, 2026), a sensitivity analysis, a trade-off assessment, and an evaluation of qualitative factors. This approach, as recommended by Ulrich and Eppinger (2011, p. 376), enabled analysis of both quantitative and qualitative aspects of the business case and provided a comprehensive representation of its complexity (Ulrich & Eppinger, 2011, p. 376). The quantitative analysis effectively evaluated measurable aspects of the business case, such as net present value (NPV), sales volume, and production costs. In contrast, the qualitative analysis served as a necessary complement, enabling an assessment of non-measurable, subjective characteristics of the BESS market, including interactions between the project and the firm, the market, and the macroeconomic environment (Ulrich & Eppinger, 2011, p. 375).

2.2.4 Survey

To evaluate the public perception of the concepts, a quantitative analysis was conducted using a survey created in Google Forms. The analysis followed the systematic approach defined by Ulrich et al. (2020, p. 173), by first defining the purpose, population, and format. Primary focus was placed on assessing whether the design concepts conveyed the intentional message or not, and further explore the actual perception of each design. The target population being the general public, further defined as people likely to passively interact with the BESS in public environments. To quantify user feedback, statements were paired with a Likert-scale to indicate their level of agreement. The final renders of each concept were used to effectively communicate the concepts to the participant, providing the necessary information. Finally, the result of the survey was summarized and analyzed to identify weaknesses and strengths of each concept.

3 Discussion

This chapter provides a critical discussion of the methods applied in the project. The chapter is based on the different phases presented in the previous section.

3.1 Ideation

This subsection critically reflects on the methods utilized during the ideation phase. It analyzes how the chosen methods supported the concept generation and highlights the limitations and obstacles encountered during the process.

3.1.1 Initial Direction

Setting the initial direction for the Volvo BESS project highlighted the contrast between theoretical approaches to product development and the complexity of reality. Ulrich et al. (2020, p. 97) emphasizes the importance of establishing measurable target specifications early in the concept development phase to define exactly what the product must do. In this project, however, it proved challenging to strictly follow this methodology.

Because of the project's strong focus on exterior design, critical metrics, such as maintaining an interior temperature of 23°C or offering specific temperature control, were heavily dependent on internal components, whose type had not been fully specified. However, it is worth noting that Ulrich et al. (2020, pp. 97–98) do claim that specifications are iterative in nature and are often revised as the team learns more about limitations and possibilities. By viewing the specification as a living document rather than a static truth, the initial deviation of not having a complete quantifiable list from the start can be viewed as a natural and necessary adaptation to the uncertainty of the project rather than as a methodological error. Rather than a complete list, the requirements list functioned more as a set of non-negotiable boundary conditions. Despite this deviation from the theoretical ideal, the requirements list proved to be very valuable later in the process during concept selection. The group integrated the core requirements as evaluation criteria in a Pugh matrix, which provided a structured way to evaluate and score the concepts against the constraints of the project.

While the requirements list addressed the functional boundaries, the mood board approach was crucial in addressing the subjective aspects of the design. Ulrich et al. (2020, pp. 218–219) points out that industrial design is essential for fulfilling aesthetic needs. Based on experience, the mood board was an invaluable communication tool in establishing the product's visual identity (Ulrich et al., 2020, pp. 222–223). Text-based descriptions of emotions can be interpreted differently by each team member. Here, the mood boards were useful for synchronizing the team's mental models and translated abstract words into a common visual language. During the concept generation phase, it served as a direct source of inspiration and helped the team evoke the intended emotions and design expressions during sketching.

The most important insight from using these two methods in parallel was the insights it provided regarding the inevitable clash between aesthetic vision and technical constraints, which is what the literature defines as trade-offs (Ulrich et al., 2020, p. 112). A clear example was the initial aesthetic desire to remove corner castings and forklift pockets to achieve a cleaner and more modern design in line with the mood board. However, the requirements list dictated that the BESS unit had to be able to be transported as a standard 20 foot container. Ultimately, the functional requirement had to take precedence over the aesthetic desire. This conflict clearly illustrated the necessary, and sometimes challenging, interactions between industrial design goals and technical specifications.

3.1.2 Brainstorming

An alternative creative method that could have been used to generate solutions within the project is traditional brainstorming. The focus of this method is to generate many ideas by encouraging creativity being through building on each other's ideas (Wikberg Nilsson et al., 2021, p. 137). Although this method avoids criticism and encourages the participants to think freely, it was considered less suitable compared to the other methods used. The Brainwriting 6-3-5 method expands upon traditional brainstorming and was more useful in this specific project since it allowed the team to build upon each others' ideas in a structured way, which ensured a more even contribution within the team.

Brainwriting 6-3-5 was effective for developing new design solutions, in terms of generating a large number of ideas and ensuring that the team worked closely together, expanding upon each other's ideas. This allowed exploration of several different design approaches, which was useful in this stage of the project. However, the initial question: "How can a functional and safe BESS system be visually integrated and pleasing to its environment?" was rather broad, and the overall process could have benefited from making the initial question even more specific. A possible solution could have been to develop ideas based on different themes or requirements, making sure that the generated ideas would be even more aligned with the overall objective. Structuring the method in this way could also contribute to further facilitation, organization, and evaluation of the generated ideas, and a clearer view on how to implement them moving forward.

The SCAMPER method was used to develop and refine the ideas generated from Brainwriting even further. This method was efficient for exploring how the different ideas could be modified and developed. Building upon the argument that the Brainwriting method could have benefited from being even more structured with a more specific initial question, the SCAMPER method contributed by introducing some of that structure by forcing a selection of useful ideas and an organized use of techniques. At the same time, since some of the ideas generated from Brainwriting were rather unrealistic, using the SCAMPER method was not productive for all ideas. However, in creative processes such as this, it is still important to avoid being overly rigid or creating restrictions, since this could risk limiting the exploration of the design space and generation of spontaneous ideas.

The combination of Brainwriting 6-3-5 and SCAMPER resulted in an efficient and structured ideation workflow where all team members were encouraged to contribute without facing criticism or judgment. The processes helped generate a wide range of ideas that were further developed within the project. However, it could be argued that the preparation and initial question could have been formulated differently to yield an even more useful result. For future work, being intentional with and carefully choosing the initial question while combining structured methods with more open ones could support both structured development and spontaneous idea generation.

3.1.3 Morphological Matrix

The Morphological Matrix proved invaluable for organizing complex ideas and maintaining an overview of the BESS project. Even though the team encountered minor challenges regarding the initial organization of the matrix and its effectiveness, the final output successfully facilitated a structured and iterative work process.

A critical decision was the rejection of the Morpheus software (Martinsson Bonde et al., 2022). While Morpheus offers benefits like compatibility notations and calculates the total number of possible solutions, the team had a number of possible solutions that far exceeded the capabilities of the program, thus making it impossible to use. Moreover, the team found it more effective to remain within Miro for a more interactive and centralized environment for collaborative working where links, drawings, and physical media could be integrated directly into the matrix.

Regarding concept synthesis, the team rejected the randomized combination generator method. With the high volume of sub-solutions, randomizing combinations would have generated excessive "noise" and impractical solutions. Instead, the team utilized a targeted strategy of defining thematic directions, such as "Unexpected" and "Playful" versus "Functional" and "Sustainable", ensuring both creative and grounded options to explore a wide design space.

The team followed Strategy Proposal 1 according to (Almefelt, 2026, pp. 50–76): deriving sub-solutions for all sub-functions and combining them into full system alternatives before evaluating and rejecting inferior concepts. This allowed for a holistic consideration of the system in one round. Conversely, Strategy Proposal 2 (step-by-step by system level) was rejected. This hierarchical approach, which requires keeping only one solution at each level before moving to the next, did not suit the BESS project. Because the sub-functions of a battery system are often interdependent, locking in a solution at a high hierarchical level prematurely would have narrowed the design space.

3.2 Evaluation

This subsection critically reflects on the methods utilized during the evaluation phase. It analyzes how the chosen methods supported the concept evaluation and highlights the limitations and obstacles encountered

during the process.

3.2.1 Pugh-matrix

The Pugh-matrix applied in the project proved to be limited in its ability to guide the elimination of concepts. While the method provided some guidance in ranking the concepts, the high number of n/a criteria caused the method to be ineffective.

The lack of information for many of the functional- and technical criteria made the matrix too rigid to handle the subjective and abstract nature of exterior design. This was caused by the fact that the majority of the criteria were not properly adapted to fit into the Pugh-matrix. The criteria used in a Pugh-matrix should be based on customer needs and the needs of the company, such as cost related criteria (Ulrich et al., 2020, p. 157). However, in this case, the selected criteria were either highly technical or excessively abstract and subjective. Due to the lack of knowledge, the high level of technicality could not be properly assessed and evaluated, and the aesthetic criteria may have been highly influenced by subjectivity towards each concept. Additionally, as previously mentioned, the project specifically focused on the exterior design. Therefore, one could debate that the Pugh-matrix should have put less emphasis on technical criteria, which could have provided a more efficient evaluation. Additionally, the original design by Volvo Energy should not have been used as the reference since it led to little or no variation between the concepts. For a better result, multiple iterations should have been made with the different concepts as a reference, for a better assessment.

While the Pugh matrix provided insight into the relative strengths and weaknesses of the concepts, no eliminations were made solely based on its results. The decision to further develop all concepts at this stage in the project could be deemed both positive and negative. It is positive in the sense that eliminating concepts too early could potentially "kill" those with high potential, and by keeping all concepts, their full potential could be further explored. However, it is negative in the sense that a higher number of concepts requires more development time and resources.

3.2.2 The Value-method

Addressing subjective criteria, particularly those related to aesthetics, Ulrich et al. (2020, p. 154) argues that a structured approach offers several benefits when evaluating concepts on subjective dimensions. When the number of concepts has been reduced to around three, it is suggested that mock-ups or models of the concepts should be developed and that target customers should be involved to assess subjective criteria (Ulrich et al., 2020, p. 161). The Value-method was effective for addressing the subjective, aesthetic criteria of the concepts within the time scope available for the evaluation phase of the project. The structured approach, which enabled equivalent evaluation of all concepts and the clear output showcasing that one concept had significantly more limitations than the others, confirms that this method was highly appropriate for assessing the subjective requirements. Moreover, one could argue that it would have been relevant to include target customers earlier in the development process and that this would have strengthened the credibility of the evaluation process and better ensured continued alignment with requirements and customer needs (Ulrich et al., 2020, p. 185). While this argument would have been highly relevant, the time scope available for the project created a need for prioritization of activities within each development phase. Consequently, the involvement of target customers was included later in the evaluation process. However, by continuously addressing the requirements list that reflected customer needs in discussions and having regular communication with the client, Volvo Energy, the evaluation process, which included the Value-method, was performed with a confidence of continued alignment with customer needs.

After concepts screening, which includes methods like the Pugh-matrix and the Value-method, Ulrich et al. (2020, p. 155) suggest using concept scoring as the next step in the development process, to evaluate the concepts using a more refined comparison with respect to each criterion. While a scoring matrix would have been beneficial for addressing some functional requirements, this method was not used in the project due to the nature of the main evaluation criteria and the project's time constraints. At this stage, three concepts were already selected for further development and the highest prioritization was to further develop the concepts rather than eliminating. Taking the project's objective and subjective evaluation criteria into account, it was concluded that assigning reliable weights to qualitative factors in a scoring matrix would risk introducing a false level of accuracy. Instead, a more holistic approach was applied, combining commercial assessment with gathering feedback from target customers. This resulted

in an adequate differentiation between the three concepts, generating more actionable, nuanced insights than a scoring-matrix would have provided at this stage.

3.2.3 Commercial Assessment

A major strength of the commercial assessment was its combination of quantitative and qualitative analysis, enabling both measurable and subjective aspects of commercial viability to be considered. This was particularly relevant since many potential benefits of improved exterior design, such as brand differentiation and perceived customer value, are difficult to quantify. However, the qualitative assessment relied largely on the project team's own interpretations and was not fully validated with customers or industry experts, which may have reduced the reliability of the conclusions.

A second limitation concerns the assumptions underlying the financial model. Since accurate data was not available for all commercial data of the financial model, key parameters such as sales volume, product pricing, development costs, and gross margins were based on informed assumptions rather than empirical market data. While such assumptions were necessary due to project constraints, they introduced uncertainty into the resulting NPV calculations. However, the sensitivity analysis partially mitigated this limitation by illustrating how variations in key parameters affected profitability.

Finally, the commercial assessment primarily evaluated a single forecasted market scenario over a five-year period. While this provided a useful estimate of potential profitability, it simplified the complexity and uncertainty of the rapidly evolving BESS market. Factors such as technological developments, changing regulations, competitor actions, and fluctuations in customer demand may significantly influence future market conditions. A more robust assessment could therefore have incorporated multiple market scenarios or alternative forecasting approaches to better account for long-term uncertainty.

3.2.4 Survey

The quantitative methodology, systematically structured around the framework provided by Ulrich et al. (2020, p. 173), yielded a foundation for further evaluation of public perception of the concepts. Utilizing the final renders provided the participants with the necessary visual context required to formally evaluate each concept. Furthermore, pairing statements with a Likert-scale, the subjective responses were successfully translated into quantifiable data points that helped identify strengths and weaknesses.

However, by evaluating public perception through a digital survey only, a variety of trade-offs and limitations had to be considered when interpreting the results. İsmailoğlu and Akkan Çavdar (2024) states that a larger sample size reduces sampling errors and the likelihood of biases to occur. Therefore, a smaller sample size could potentially provide a deceptive conclusion, as the data gathered would not accurately represent the entire target population. While a survey was highly effective when collecting quantitative data in a short time span, the feedback merely represented an isolated, deliberate assessment rather than an actual real-world interaction. Therefore, complementing the survey with real-world observations could have provided a more comprehensive and nuanced understanding of public perception.

Furthermore, relying exclusively on Likert-scale agreement levels entailed analytical challenges, particularly regarding the nuance of design feedback. Quantifying aesthetic perception is a complex task, and while the Likert-scale could confirm that the concepts successfully communicated their intended messages, it did not account for the underlying reasons. As a result, the survey was only able to provide *what* the participant perceived, but not *why* it was perceived that way. Combining the survey with a deeper qualitative analysis could have provided a more accurate understanding by further exploring the subject deeper.

4 Conclusion

This project navigated the intersection between aesthetic vision and functional constraints. The design process, including ideation, evaluation, and testing, gave the project an overall structure while keeping consistent focus on the project aim.

An analysis of the design methodology reveals that using both creative and systematic methods for generating design solutions created a process that enabled a wider design space to be explored. The creative methods, Brainstorming 6-3-5 and SCAMPER, provided a structured framework while encouraging the generation of a high volume of ideas, even though they, in retrospect, would have benefited from having even more specific guidelines and starting points. The Morphological Matrix with randomized combinations was shown to be highly valuable for organizing a high volume of sub-functions and therefore, an organizing strategy based on thematic directions produced more valuable results in the project. The Pugh matrix showcased limitations due to insufficient information on functional and technical criteria, which highlights the complexity of using traditional evaluation methods to evaluate subjective- and design related criteria. This emphasizes the importance of using qualitative methods such as the Value method as a complement, to enable sufficient exploration and evaluation of semantic values. Consequently, while the survey successfully provided valuable and quantitative data, future analysis would benefit from deeper qualitative methods to uncover the specific underlying factors driving public perception

The project exposed the inevitable trade-offs between aesthetic vision and functional constraints. This required creating a process in which the contrast between a modern, clean appearance and the functional requirements of the BESS units, including specific dimensions and features, was made explicit and subsequently synthesized. The final selection of three concepts yielded a diverse range of exterior solutions, reflecting Volvo Energy's core values. By re-imagining and re-designing the BESS units in this way, the project contributes to the aim of creating a modern-looking energy solution that represents the Volvo brand.

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